Integration

Chapter Overview — Integration Methods

Solution 2

In this chapter you will integrate a greater variety of expressions and solve differential equations.

Standard results

Know the antiderivatives of standard functions.

$$\int \sec^2 x \, dx = \tan x + C.$$

Substitution

Introduce a new variable to make an integral easier.

$$\int x\sqrt{2x+5}\, dx$$

$$u = 2x+5, \quad x = \frac{u-5}{2}, \quad dx = \frac{1}{2}\, du.$$

$$\int x\sqrt{2x+5}\, dx = \frac{1}{4}\int (u-5)\sqrt{u}\, du.$$

Reverse chain rule

Think about what differentiates to give the expression.

$$\int \cos 4x \, dx = \frac{1}{4}\sin 4x + C.$$

$$\int \sin^3 x \cos x \, dx = \frac{1}{4}\sin^4 x + C.$$

Integration by parts

Integrate a product using the product rule in reverse.

$$\int x \cos x \, dx$$

$$u = x, \quad \frac{dv}{dx} = \cos x, \quad \frac{du}{dx} = 1, \quad v = \sin x.$$

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx$$

$$= x \sin x + \cos x + C.$$

Chapter Overview — Further Applications

Solution 3

Integrating partial fractions

Split a rational expression into partial fractions so that each term is easier to integrate.

$$\int \frac{3x+5}{(x+1)(x+2)}\,dx$$

Solving differential equations

Find one variable in terms of another without derivatives remaining. For $V > 0$:

$$\frac{dV}{dt} = -kV$$

$$\int \frac{1}{V}\,dV = \int -k\,dt$$

$$\ln V = -kt + C$$

$$V = e^{-kt+C} = Ae^{-kt}.$$

Approximating areas: the trapezium rule

Instead of integrating, split the area under the graph into trapeziums and add their areas to approximate the total.

Integrating Standard Functions

Solution 4

$\int x^n\,dx =$	$\frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\int e^x\,dx =$	$e^x + C$
$\int \frac{1}{x}\,dx =$	$\ln\lvert x\rvert + C$ The absolute value allows negative x; $x = 0$ is excluded.
$\int \cos x\,dx =$	$\sin x + C$ Picture sine above cosine: down is differentiating; up is integrating. Change the sign when travelling the other way.
$\int \sin x\,dx =$	$-\cos x + C$
$\int \sec^2 x\,dx =$	$\tan x + C$ Memorise $\int \sec^2 x\,dx = \tan x + C$.
$\int \operatorname{cosec} x \cot x\,dx =$	$-\operatorname{cosec} x + C$
$\int \operatorname{cosec}^2 x\,dx =$	$-\cot x + C$
$\int \sec x \tan x\,dx =$	$\sec x + C$

Quickfire Questions

Solution 5

$$\int \sec x \tan x \, dx =$$

$\sec x + C$

$$\int \sin x \, dx =$$

$-\cos x + C$

$$\int \operatorname{cosec}^2 x \, dx =$$

$-\cot x + C$

$$\int -\cos x \, dx =$$

$-\sin x + C$

Quickfire Questions

Solution 6

$$\int \sec^2 x \, dx =$$

$\tan x + C$

$$\int \operatorname{cosec} x \cot x \, dx =$$

$-\operatorname{cosec} x + C$

$$\int 1/x \, dx =$$

$\ln |x| + C$

$$\int -\sin x \, dx =$$

$\cos x + C$

Quickfire Questions — Without Looking Back

Solution 7

$\int \operatorname{cosec}^2 x \, dx =$ $-\cot x + C$

$\int \sin x \, dx =$ $-\cos x + C$

$\int \sec x \tan x \, dx =$ $\sec x + C$

$\int \cos x \, dx =$ $\sin x + C$

Test Your Understanding — Standard Integrals

Solution 8

1. Find $\int (2\cos x + 3/x - \sqrt{x})\, dx$.

1

$$2\sin x + 3\ln|x| - \frac{2}{3}x^{3/2} + C.$$

2. Find $\int \frac{\cos x}{\sin^2 x}\, dx$.

Hint: Which reciprocal trigonometric functions does this simplify to?

2

$$\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C.$$

3. Given $\int_a^{3a} \frac{2x+1}{x}\, dx = \ln 12$, find the exact value of a.

3

$$\int_a^{3a} \left(2 + \frac{1}{x}\right) dx = [2x + \ln|x|]_a^{3a}$$
$$= (6a + \ln|3a|) - (2a + \ln|a|)$$
$$= 4a + \ln 3 = \ln 12.$$
$$4a = \ln 12 - \ln 3 = \ln 4 \quad \Rightarrow \quad a = \frac{1}{4}\ln 4.$$

Note

You may split a numerator:

$$\frac{x+1}{x} = \frac{x}{x} + \frac{1}{x} = 1 + \frac{1}{x}.$$

You cannot split a denominator:

$$\frac{x}{x+1} \neq \frac{x}{x} + \frac{x}{1}.$$

4. Given that $a > 0$ and $\int_a^{2a} \frac{3x-1}{x}\,dx = 6 + \ln(\frac{1}{2})$, find the exact value of a.

4

$$\int_a^{2a} \frac{3x-1}{x}\,dx = \int_a^{2a} \left(3 - \frac{1}{x}\right) dx$$
$$= [3x - \ln|x|]_a^{2a}$$
$$= (6a - \ln 2a) - (3a - \ln a)$$
$$= 3a - \ln 2a + \ln a$$
$$= 3a - (\ln 2 + \ln a) + \ln a$$
$$= 3a - \ln 2 = 3a + \ln(\tfrac{1}{2}).$$
$$3a + \ln(\tfrac{1}{2}) = 6 + \ln(\tfrac{1}{2})$$
$$a = 2.$$

5. Given that $a > 0$ and $\int_{\ln 1}^{\ln a} (e^x + e^{-x})\,dx = 48/7$, find the exact value of a.

5

$$\int_{\ln 1}^{\ln a} (e^x + e^{-x})\,dx = [e^x - e^{-x}]_{\ln 1}^{\ln a}$$
$$= (e^{\ln a} - e^{-\ln a}) - (e^{\ln 1} - e^{-\ln 1})$$
$$= \left(a - \frac{1}{a}\right) - (1 - 1).$$
$$a - \frac{1}{a} = \frac{48}{7}$$
$$7a^2 - 48a - 7 = 0$$
$$(7a + 1)(a - 7) = 0.$$
$$a = 7 \quad \text{since } a > 0.$$

7. $f(x) = \frac{1}{8}x^{3/2} - 4/x$, $x > 0$.

a. Solve $f(x) = 0$.

b. Find $\int f(x)\,dx$.

c. Evaluate $\int_1^4 f(x)\,dx$, giving your answer as $p + q\ln r$, where p, q, r are rational numbers.

7a

$$\tfrac{1}{8}x^{3/2} - \frac{4}{x} = 0$$
$$\tfrac{1}{8}x^{5/2} - 4 = 0 \quad (x \neq 0)$$
$$x^{5/2} = 32 \quad \Rightarrow \quad x = 4.$$

7b

$$\int \left(\tfrac{1}{8}x^{3/2} - \frac{4}{x}\right) dx$$
$$= \frac{x^{5/2}}{20} - 4\ln|x| + C.$$

7c

$$\int_1^4 f(x)\,dx = \left[\frac{x^{5/2}}{20} - 4\ln|x|\right]_1^4$$
$$= \left(\frac{32}{20} - 4\ln 4\right) - \left(\frac{1}{20} - 4\ln 1\right)$$
$$= \frac{31}{20} - 4\ln 4.$$

Integrating a Linear Inner Function

Solution 11

$\frac{d}{dx}\sin(3x+1) =$

$3\cos(3x+1).$

Therefore $\int \cos(3x+1)\,dx =$

$\frac{1}{3}\sin(3x+1) + C.$

For a linear inner function $ax+b$, integrate as before and divide by a, where $a \neq 0$:

$$\int f'(ax+b)\,dx = \frac{1}{a}f(ax+b) + C.$$

1. $\int e^{3x}\,dx$

[1] $\frac{1}{3}e^{3x} + C$

2. $\int \frac{1}{5x+2}\,dx$

[2] $\frac{1}{5}\ln|5x+2| + C$

3. $\int 2\sec^2(3x-2)\,dx$

[3] $\frac{2}{3}\tan(3x-2) + C$

4. $\int (3x+4)^3\,dx$

[4] $\frac{1}{12}(3x+4)^4 + C$

5. $\int \sin(1-5x)\,dx$

[5] $\frac{1}{5}\cos(1-5x) + C$

6. $\int \frac{1}{3(4x-2)^2}\,dx$

[6] $-\frac{1}{12}(4x-2)^{-1} + C$

7. $\int (10x+11)^{12}\,dx$

[7] $\frac{1}{130}(10x+11)^{13} + C$

Tip

For $\int (ax+b)^n\,dx$, divide by both $n+1$ and a ($n \neq -1$, $a \neq 0$).

Check Your Understanding — Linear Inner Functions

Solution 12

$\int e^{3x+1}\,dx =$ $\frac{1}{3}e^{3x+1} + C$

$\int \frac{1}{1-2x}\,dx =$ $-\frac{1}{2}\ln|1-2x| + C$

$\int (4-3x)^5\,dx =$ $-\frac{1}{18}(4-3x)^6 + C$

$\int \sec(3x)\tan(3x)\,dx =$ $\frac{1}{3}\sec(3x) + C$

Exercise 11B — Pages 297–298 Solution 13

Find the indefinite integral of each expression with respect to x.

1a. $\sin(2x+1)$

1a $-\frac{1}{2}\cos(2x+1)+C$

1c. $4e^{x+5}$

1c $4e^{x+5}+C$

1e. $\operatorname{cosec}^2(3x)$

1e $-\frac{1}{3}\cot 3x+C$

1f. $\sec 4x \tan 4x$

1f $\frac{1}{4}\sec 4x+C$

1g. $3\sin(\frac{1}{2}x+1)$

1g $-6\cos(\frac{1}{2}x+1)+C$

1h. $\operatorname{cosec} 2x \cot 2x$

1h $-\frac{1}{2}\operatorname{cosec} 2x+C$

2a. $e^{2x}-\frac{1}{2}\sin(2x-1)$

2a $\frac{1}{2}e^{2x}+\frac{1}{4}\cos(2x-1)+C$

2b. $(e^x+1)^2$

2b $\frac{1}{2}e^{2x}+2e^x+x+C$

2c. $\sec^2(2x)(1+\sin 2x)$

2c $\frac{1}{2}\tan 2x+\frac{1}{2}\sec 2x+C$

2d. $\frac{3-2\cos(\frac{1}{2}x)}{\sin^2(\frac{1}{2}x)}$

2d $-6\cot(\frac{1}{2}x)+4\operatorname{cosec}(\frac{1}{2}x)+C$

2e. $e^{3-x}+\sin(3-x)+\cos(3-x)$

2e $-e^{3-x}+\cos(3-x)-\sin(3-x)+C$

3a. $\frac{1}{2x+1}$

3a $\frac{1}{2}\ln|2x+1|+C$

3b. $\frac{1}{(2x+1)^2}$

3b $-\frac{1}{2(2x+1)}+C$

3c. $(2x+1)^2$

3c $\frac{1}{6}(2x+1)^3+C$

3d. $\frac{3}{4x-1}$

3d $\frac{3}{4}\ln|4x-1|+C$

3f. $\frac{3}{(1-4x)^2}$

3f $\frac{3}{4(1-4x)}+C$

3h. $\frac{3}{(1-2x)^3}$

3h $\frac{3}{4(1-2x)^2}+C$

3j. $\frac{5}{3-2x}$

3j $-\frac{5}{2}\ln|3-2x|+C$

4a. $3\sin(2x+1)+\frac{4}{2x+1}$

4a $-\frac{3}{2}\cos(2x+1)+2\ln|2x+1|+C$

4c. $\frac{1}{\sin^2 2x}+\frac{1}{1+2x}+\frac{1}{(1+2x)^2}$

4c $-\frac{1}{2}\cot 2x+\frac{1}{2}\ln|1+2x|-\frac{1}{2(1+2x)}+C$

Exercise 11B — Pages 297–298 Solution 14

6. Given $\int_3^b (2x-6)^2\,dx = 36$, find b.

6

$$\int_3^b (2x-6)^2\,dx = \int_3^b (4x^2-24x+36)\,dx$$

$$= \left[\frac{4x^3}{3}-12x^2+36x\right]_3^b = 36.$$

$$\left(\frac{4b^3}{3}-12b^2+36b\right)-(36-108+108)=36$$

$$\frac{4b^3}{3}-12b^2+36b-72=0$$

$$b^3-9b^2+27b-54=0$$

$$(b-6)(b^2-3b+9)=0.$$

$b=6$ since $b^2-3b+9>0$.

7. Given $\int_{e^2}^{e^8} \frac{1}{kx}\,dx = \frac{1}{4}$, find k.

7

$$\int_{e^2}^{e^8} \frac{1}{kx}\,dx = \left[\frac{1}{k}\ln x\right]_{e^2}^{e^8} = \frac{1}{4}$$

$$\frac{8}{k}-\frac{2}{k}=\frac{1}{4}$$

$$k=32-8=24.$$

Exercise 11B — Pages 297–298 Solution 15

8. Given $\int_{\pi/(4k)}^{\pi/(3k)} (1 - \pi \sin kx)\,dx = \pi(7 - 6\sqrt{2})$, find the exact value of k.

8

$$\int_{\pi/(4k)}^{\pi/(3k)} (1 - \pi \sin kx)\,dx = \left[x + \frac{\pi}{k}\cos kx\right]_{\pi/(4k)}^{\pi/(3k)}$$

$$= \left(\frac{\pi}{3k} + \frac{\pi}{k}\cos\frac{\pi}{3}\right) - \left(\frac{\pi}{4k} + \frac{\pi}{k}\cos\frac{\pi}{4}\right)$$

$$= \left(\frac{\pi}{3k} + \frac{\pi}{2k}\right) - \left(\frac{\pi}{4k} + \frac{\pi}{\sqrt{2}k}\right)$$

$$= \frac{\pi}{k}\left[\left(\frac{1}{3} + \frac{1}{2}\right) - \left(\frac{1}{4} + \frac{1}{\sqrt{2}}\right)\right]$$

$$= \frac{\pi}{k}\left(\frac{7}{12} - \frac{\sqrt{2}}{2}\right) = \pi(7 - 6\sqrt{2})$$

$$\frac{\pi}{k}\left(\frac{7 - 6\sqrt{2}}{12}\right) = \pi(7 - 6\sqrt{2})$$

$$k = \frac{1}{12}.$$

Challenge — Exercise 11B Solution 16

Challenge

Given $\int_5^{11} \frac{1}{ax+b}\,dx = \frac{1}{a}\ln(41/17)$, where a, b are integers and $0 < a < 10$, find two different pairs of values of a, b.

Working

An antiderivative is $\frac{1}{a}\ln|ax+b| + \frac{1}{a}\ln k$, where $k > 0$ is a constant.

$$\left[\frac{1}{a}\ln(k|ax+b|)\right]_5^{11}$$

$$= \frac{1}{a}[\ln(k|11a+b|) - \ln(k|5a+b|)].$$

The constant cancels, giving

$$\ln\left|\frac{11a+b}{5a+b}\right| = \ln\frac{41}{17}.$$

$$\frac{11a+b}{5a+b} = \pm\frac{41}{17}.$$

Case 1

Case 1

$\frac{11a+b}{5a+b} = \frac{41}{17}$.

$187a + 17b = 205a + 41b$

$18a = -24b \quad \Rightarrow \quad 3a = -4b.$

Thus a is a multiple of 4 between 0 and 10:

$a = 4 \Rightarrow b = -3; \qquad a = 8 \Rightarrow b = -6.$

Case 2

Case 2

$\frac{11a+b}{5a+b} = -\frac{41}{17}$.

$187a + 17b = -205a - 41b$

$392a = -58b \quad \Rightarrow \quad b = -\frac{196}{29}a.$

This is not an integer for $a = 1, \ldots, 9$. It would also put a pole inside the interval, so cannot define the given integral.

The only pairs are $(4, -3)$ and $(8, -6)$; their denominators stay positive on $[5, 11]$.

Integrating Using Trigonometric Identities

Solution 17

Some expressions, including $\sin^2 x$ and $\sin x \cos x$, need a trigonometric identity before integration.

Find $\int \sin^2 x \, dx$.

Which identity contains $\sin^2 x$?

Sine squared

$$\cos 2x = 1 - 2\sin^2 x$$

$$\sin^2 x = \tfrac{1}{2} - \tfrac{1}{2}\cos 2x$$

$$\int \sin^2 x \, dx = \tfrac{1}{2}x - \tfrac{1}{4}\sin 2x + C.$$

Find $\int \cos^2 x \, dx$.

Cosine squared

$$\cos 2x = 2\cos^2 x - 1$$

$$\cos^2 x = \tfrac{1}{2} + \tfrac{1}{2}\cos 2x$$

$$\int \cos^2 x \, dx = \tfrac{1}{2}x + \tfrac{1}{4}\sin 2x + C.$$

Find $\int \sin 3x \cos 3x \, dx$.

Product

$$\sin 6x = 2\sin 3x \cos 3x$$

$$\sin 3x \cos 3x = \tfrac{1}{2}\sin 6x$$

$$\int \sin 3x \cos 3x \, dx = -\tfrac{1}{12}\cos 6x + C.$$

Find $\int \tan^2 x \, dx$.

Tangent squared

Recall $1 + \tan^2 x = \sec^2 x$.

$$\int \tan^2 x \, dx = \int (\sec^2 x - 1)\, dx$$

$$= \tan x - x + C.$$

Check Your Understanding — Trigonometric Identities

Solution 18

Find $\int (\sec x + \tan x)^2 \, dx$.

$$\int (\sec x + \tan x)^2 \, dx$$

$$= \int (\sec^2 x + 2\sec x \tan x + \tan^2 x)\, dx$$

$$= \tan x + 2\sec x + \tan x - x + C$$

$$= 2\tan x + 2\sec x - x + C.$$

Find $\int_{\pi/6}^{\pi/3} \sin^2(3x)\, dx$.

$$\cos 6x = 1 - 2\sin^2(3x)$$

$$\sin^2(3x) = \tfrac{1}{2} - \tfrac{1}{2}\cos 6x.$$

$$\int_{\pi/6}^{\pi/3} \sin^2(3x)\, dx = \int_{\pi/6}^{\pi/3} (\tfrac{1}{2} - \tfrac{1}{2}\cos 6x)\, dx$$

$$= \left[\tfrac{1}{2}x - \tfrac{1}{12}\sin 6x\right]_{\pi/6}^{\pi/3}$$

$$= \left(\frac{\pi}{6} - \frac{1}{12}\sin 2\pi\right) - \left(\frac{\pi}{12} - \frac{1}{12}\sin \pi\right)$$

$$= \frac{\pi}{12}.$$

Exercise 11C — Trigonometric Identities

Solution 19

3. Show that $\int_{\pi/4}^{\pi/2} \sin^2 x\,dx = \frac{2+\pi}{8}$.

3

$$\int_{\pi/4}^{\pi/2} \sin^2 x\,dx = \frac{1}{2}\int_{\pi/4}^{\pi/2} (1-\cos 2x)\,dx$$
$$= \frac{1}{2}\left[x - \frac{1}{2}\sin 2x\right]_{\pi/4}^{\pi/2}$$
$$= \frac{1}{2}\left[\left(\frac{\pi}{2} - 0\right) - \left(\frac{\pi}{4} - \frac{1}{2}\right)\right]$$
$$= \frac{1}{2}\left(\frac{\pi}{4} + \frac{1}{2}\right) = \frac{2+\pi}{8}.$$

5a. Expand $\sin(3x+2x)$ and $\sin(3x-2x)$, using the compound-angle formulae. Hence show that $\sin 5x + \sin x = 2\sin 3x\cos 2x$.

5a

$\sin(3x+2x) = \sin 3x\cos 2x + \cos 3x\sin 2x$

$\sin(3x-2x) = \sin 3x\cos 2x - \cos 3x\sin 2x.$

Adding gives $\sin 5x + \sin x = 2\sin 3x\cos 2x.$

5b. Hence find $\int \sin 3x\cos 2x\,dx$.

5b

$$\int \sin 3x\cos 2x\,dx = \frac{1}{2}\int (\sin 5x + \sin x)\,dx$$
$$= \frac{1}{2}\left(-\frac{1}{5}\cos 5x - \cos x\right) + C$$
$$= -\frac{1}{10}\cos 5x - \frac{1}{2}\cos x + C.$$

Exercise 11C — Trigonometric Identities

Solution 20

6. $f(x) = 5\sin^2 x + 7\cos^2 x$.

a. Show that $f(x) = \cos 2x + 6$.

6a

$$f(x) = 5\sin^2 x + 7 - 7\sin^2 x = 7 - 2\sin^2 x$$
$$= 7 - 2\left[\tfrac{1}{2}(1-\cos 2x)\right]$$
$$= 7 - 1 + \cos 2x = \cos 2x + 6.$$

6b. Find $\int_0^{\pi/4} f(x)\,dx$.

6b

$$\int_0^{\pi/4} f(x)\,dx = \int_0^{\pi/4} (\cos 2x + 6)\,dx$$
$$= \left[\tfrac{1}{2}\sin 2x + 6x\right]_0^{\pi/4}$$
$$= \tfrac{1}{2} + \tfrac{3\pi}{2} = \tfrac{1}{2}(1+3\pi).$$

7a. Show that $\cos^4 x = \frac{1}{8}\cos 4x + \frac{1}{2}\cos 2x + \frac{3}{8}$.

7a

$$\cos^4 x = (\cos^2 x)^2 = \left[\tfrac{1}{2}(\cos 2x + 1)\right]^2$$
$$= \tfrac{1}{4}(\cos^2 2x + 2\cos 2x + 1)$$
$$= \tfrac{1}{4}\left[\tfrac{1}{2}(\cos 4x + 1) + 2\cos 2x + 1\right]$$
$$= \tfrac{1}{8}\cos 4x + \tfrac{1}{2}\cos 2x + \tfrac{3}{8}.$$

7b. Hence find $\int \cos^4 x\,dx$.

7b

$$\int \cos^4 x\,dx = \int \left(\tfrac{1}{8}\cos 4x + \tfrac{1}{2}\cos 2x + \tfrac{3}{8}\right)dx$$
$$= \tfrac{1}{32}\sin 4x + \tfrac{1}{4}\sin 2x + \tfrac{3}{8}x + C.$$

Reverse Chain Rule — Consider Then Scale

Solution 21

Complicated expressions can look like the result of applying the chain rule. Use "consider then scale".

1. Consider an expression whose derivative will look similar.
2. Differentiate and adjust for any scale difference.

$\int x(x^2+5)^3\,dx$

The first x looks as though it arose from differentiating x^2 inside the brackets.

1

Consider $y=(x^2+5)^4$.

$\dfrac{dy}{dx}=4(x^2+5)^3\times 2x$

$=8x(x^2+5)^3$.

Thus

$\int x(x^2+5)^3\,dx$

$=\dfrac{1}{8}(x^2+5)^4+C$.

$\int \cos x\sin^2 x\,dx$

The $\cos x$ probably arose from differentiating $\sin x$.

2

Consider $y=\sin^3 x$.

$\dfrac{dy}{dx}=3\sin^2 x\cos x$.

Thus

$\int \cos x\sin^2 x\,dx$

$=\dfrac{1}{3}\sin^3 x+C$.

$\int \dfrac{2x}{x^2+1}\,dx$

The $2x$ probably arose from differentiating x^2.

Consider $y=\ln|x^2+1|$.

$\dfrac{dy}{dx}=\dfrac{2x}{x^2+1}$.

Thus

$\int \dfrac{2x}{x^2+1}\,dx$

$=\ln|x^2+1|+C$.

Reverse Chain Rule — Common Patterns

Solution 22

Integration by inspection / reverse chain rule

Consider an expression that would differentiate to the expression given. Then scale appropriately.

Common patterns:

$\int k\dfrac{f'(x)}{f(x)}\,dx$: try $\ln|f(x)|$.

$\int kf'(x)[f(x)]^n\,dx$: try $[f(x)]^{n+1}$ $(n\neq -1)$.

If the bottom of a fraction differentiates to give the top, apart from a constant scale factor, try the logarithm of the absolute value of the bottom.

$\int \dfrac{x^2}{x^3+1}\,dx$

Consider $y=\ln|x^3+1|$.

$\dfrac{dy}{dx}=\dfrac{3x^2}{x^3+1}$.

Therefore

$\int \dfrac{x^2}{x^3+1}\,dx=\dfrac{1}{3}\ln|x^3+1|+C$.

$\int xe^{x^2+1}\,dx$

Consider $y=e^{x^2+1}$.

$\dfrac{dy}{dx}=2xe^{x^2+1}$.

Therefore

$\int xe^{x^2+1}\,dx=\dfrac{1}{2}e^{x^2+1}+C$.

Quickfire — Reverse Chain Rule

Solution 23

In your head!

$\int \frac{4x^3}{x^4-1}\,dx =$

$\ln|x^4-1|+C$

$\int \cos x(\sin x-5)^7\,dx =$

$\frac{1}{8}(\sin x-5)^8+C$

$\int \frac{\cos x}{\sin x+2}\,dx =$

$\ln|\sin x+2|+C$

$\int x^2(x^3+5)^7\,dx =$

$\frac{1}{24}(x^3+5)^8+C$

$\int \cos x\, e^{\sin x}\,dx =$

$e^{\sin x}+C$

Not in your head...

$\int \frac{x}{(x^2+5)^3}\,dx$

$\int x(x^2+5)^{-3}\,dx$

$= -\frac{1}{4}(x^2+5)^{-2}+C.$

Tip

If the whole denominator is raised to a power, do not apply the logarithm pattern directly. Rewrite the expression as a product, such as $x(x^2+5)^{-3}$, then use the power pattern.

Powers of Sine and Secant

Solution 24

When we differentiate $\sin^5 x$, the power decreases:

$$\frac{d}{dx}(\sin^5 x)=\frac{d}{dx}[(\sin x)^5]=5\sin^4 x\cos x.$$

However, when we differentiate $\sec^5 x$:

$$\frac{d}{dx}[(\sec x)^5]=5\sec^4 x\times\sec x\tan x=5\sec^5 x\tan x.$$

The power of $\sec x$ does not decrease. Keep this in mind when integrating.

Find $\int \sec^4 x\tan x\,dx$.

Integral

Consider $y=\sec^4 x$.

$$\frac{dy}{dx}=4\sec^3 x\times\sec x\tan x=4\sec^4 x\tan x.$$

$$\therefore\quad \int \sec^4 x\tan x\,dx=\frac{1}{4}\sec^4 x+C.$$

Test Your Understanding — Reverse Chain Rule

Solution 25

$\int \sin x(\cos x + 1)^5 \, dx$

Consider $y = (\cos x + 1)^6$.

$\frac{dy}{dx} = -6 \sin x(\cos x + 1)^5$.

Therefore

$\int \sin x(\cos x + 1)^5 \, dx$

$= -\frac{1}{6}(\cos x + 1)^6 + C.$

$\int \frac{\sec^2 2x}{\tan 2x + 1} \, dx$

$= \frac{1}{2} \ln|\tan 2x + 1| + C.$

$\int \frac{\operatorname{cosec}^2 x}{(2 + \cot x)^3} \, dx$

2

Consider $y = (2 + \cot x)^{-2}$.

$\frac{dy}{dx} = -2(2 + \cot x)^{-3}$

$\times(-\operatorname{cosec}^2 x)$

$= 2 \operatorname{cosec}^2 x(2 + \cot x)^{-3}$.

Thus

$\int \frac{\operatorname{cosec}^2 x}{(2 + \cot x)^3} \, dx$

$= \frac{1}{2}(2 + \cot x)^{-2} + C.$

$\int x(x^2 + 2)^3 \, dx$

$= \frac{1}{8}(x^2 + 2)^4 + C.$

$\int 5 \tan x \sec^2 x \, dx$

Try $y = \sec^2 x$.

$\frac{dy}{dx} = 2 \sec x \times \sec x \tan x$

$= 2 \sec^2 x \tan x.$

$\therefore \int 5 \tan x \sec^2 x \, dx$

$= \frac{5}{2} \sec^2 x + C.$

Exercise 11D — Pages 302–303

Solution 26

4. Given that $\int_0^k kx^2 e^{x^3} \, dx = \frac{2}{3}(e^8 - 1)$, find k.

4

Let $I = \int_0^k kx^2 e^{x^3} \, dx$.

Consider $y = e^{x^3}$: $\frac{dy}{dx} = 3x^2 e^{x^3}$.

$I = \left[\frac{k}{3} e^{x^3}\right]_0^k = \frac{k}{3}(e^{k^3} - 1) = \frac{2}{3}(e^8 - 1).$

The intended positive solution is $k = 2$.

As written, the question also has one negative solution: $k \approx -5959.915974$.

To see this, put $t = -k > 0$. Then $t(1 - e^{-t^3}) = 2(e^8 - 1)$; the left side increases continuously from 0 to infinity.

5. Given that $\int_0^\theta 4 \sin 2x \cos^4 2x \, dx = \frac{4}{5}$, where $0 < \theta < \pi$, find the exact value of θ.

5

Let $I = \int_0^\theta 4 \sin 2x \cos^4 2x \, dx$.

Consider $y = \cos^5 2x$:

$\frac{dy}{dx} = -10 \sin 2x \cos^4 2x.$

$I = \left[-\frac{2}{5} \cos^5 2x\right]_0^\theta$

$= -\frac{2}{5} \cos^5 2\theta + \frac{2}{5} = \frac{4}{5}.$

$\cos^5 2\theta = -1 \quad \Rightarrow \quad \cos 2\theta = -1.$

$2\theta = \pi \quad \Rightarrow \quad \theta = \frac{\pi}{2}.$

Exercise 11D — Pages 302–303 Solution 27

6a. By writing $\cot x = \cos x / \sin x$, find $\int \cot x \, dx$.

6a

$$\int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx.$$

Consider $y = \ln|\sin x|$.

$$\frac{dy}{dx} = \frac{\cos x}{\sin x}.$$

$$\therefore \int \cot x \, dx = \ln|\sin x| + C.$$

6b. Show that $\int \tan x \, dx = \ln|\sec x| + C$.

6b

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx.$$

Consider $y = \ln|\cos x|$: $\frac{dy}{dx} = -\frac{\sin x}{\cos x}$.

$$\int \tan x \, dx = -\ln|\cos x| + C$$

$$= \ln\left|\frac{1}{\cos x}\right| + C = \ln|\sec x| + C.$$

Extension — STEP I 2013, Q4

i. Show that, for $n > 0$,

$$\int_0^{\pi/4} \tan^n x \sec^2 x \, dx = \frac{1}{n+1}$$

and $$\int_0^{\pi/4} \sec^n x \tan x \, dx = \frac{(\sqrt{2})^n - 1}{n}.$$

ii. Evaluate

$$\int_0^{\pi/4} x \sec^4 x \tan x \, dx$$

and $$\int_0^{\pi/4} x^2 \sec^2 x \tan x \, dx.$$

Extension ii

$$\frac{\pi}{4} - \frac{1}{3} \quad \text{and} \quad \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2.$$

Integration by Substitution Solution 28

A substitution can turn a complicated integral into an equivalent, simpler integral. The following is not a direct reverse-chain-rule pattern.

Use $u = 2x + 5$ to find $\int x\sqrt{2x+5}\, dx$.

Remove every reference to x, replacing it with u. Work out expressions for x and dx.

Step 1

Work out x and dx, or a useful variant.

$$\frac{du}{dx} = 2 \quad \Rightarrow \quad dx = \frac{1}{2}\, du.$$

$$x = \frac{u-5}{2}.$$

Tip: remember to reciprocate when rearranging for dx.

Step 2

Substitute into the integral.

$$\int x\sqrt{2x+5}\, dx = \int \frac{u-5}{2}\sqrt{u}\,\frac{1}{2}\, du$$

$$= \frac{1}{4}\int \sqrt{u}(u-5)\, du = \frac{1}{4}\int (u^{3/2} - 5u^{1/2})\, du.$$

Tip: take constant factors outside the integral.

Step 3

Integrate the simplified expression.

$$\frac{1}{4}\left(\frac{2}{5}u^{5/2} - \frac{10}{3}u^{3/2}\right) + C.$$

Step 4

Write the answer in terms of x.

$$\frac{(2x+5)^{5/2}}{10} - \frac{5(2x+5)^{3/2}}{6} + C.$$

Choosing a Substitution

Solution 29

Edexcel questions usually give the substitution. Other examination boards and STEP may not.

There is no fixed rule: look for expressions inside roots, powers or the denominator of a fraction. Suggest a sensible substitution for each integral.

$\displaystyle\int \cos x\sqrt{1+\sin x}\,dx$

$u = 1 + \sin x.$

$\displaystyle\int 6xe^{x^2}\,dx$

$u = x^2.$

This can also be integrated by inspection.

$\displaystyle\int \frac{xe^x}{1+x}\,dx$

$u = 1 + x.$

$\displaystyle\int e^{(1-x)/(1+x)}\,dx$

$u = \dfrac{1-x}{1+x}.$

Examples 3 and 4 practise choosing a substitution only. Their antiderivatives are not elementary, so these substitutions do not complete an integration using this chapter's methods.

Another Substitution Example

Solution 30

Use $u = \sin x + 1$ to find $\int \cos x \sin x(1+\sin x)^3\,dx$.

Step 1

Work out x and dx, or useful variants.

$u = \sin x + 1$

$\dfrac{du}{dx} = \cos x \quad \Rightarrow \quad du = \cos x\,dx.$

$\sin x = u - 1.$

We did not need x or dx separately. One could use an appropriate inverse-sine branch (for example $x = \arcsin(u-1)$), but the simplification would be more awkward.

Step 2

Substitute into the integral.

$$\int \cos x \sin x(1+\sin x)^3\,dx$$
$$= \int (u-1)u^3\,du$$
$$= \int (u^4 - u^3)\,du.$$

Step 3

Integrate.

$\dfrac{1}{5}u^5 - \dfrac{1}{4}u^4 + C.$

Step 4

Write the answer in terms of x.

$\dfrac{1}{5}(\sin x + 1)^5 - \dfrac{1}{4}(\sin x + 1)^4 + C.$

Substitution with Implicit Differentiation

Solution 31

When a square root is involved, a substitution of the form $u^2 = \cdots$ can keep the algebra tidy.

Use $u^2 = 2x + 5$, taking $u = \sqrt{2x+5} \geq 0$, to find $\int x\sqrt{2x+5}\,dx$.

Step 1

Find x and dx.

$$2u\frac{du}{dx} = 2 \quad \Rightarrow \quad dx = u\,du.$$

$$x = \frac{u^2 - 5}{2}.$$

Step 2

Substitute.

$$\int x\sqrt{2x+5}\,dx = \int \frac{u^2-5}{2}\,u \times u\,du$$

$$= \int \left(\frac{1}{2}u^4 - \frac{5}{2}u^2\right) du.$$

Step 3

Integrate.

$$\frac{1}{10}u^5 - \frac{5}{6}u^3 + C.$$

Step 4

Return to x.

$$\frac{(2x+5)^{5/2}}{10} - \frac{5(2x+5)^{3/2}}{6} + C.$$

This avoids the fractional powers that arose when using $u = 2x + 5$.

Test Your Understanding — Substitution

Solution 32

Edexcel C4 January 2012, Q6c

Using $u = 1 + \cos x$, or otherwise, show that

$$\int \frac{2\sin 2x}{1+\cos x}\,dx = 4\ln(1+\cos x) - 4\cos x + k,$$

where k is a constant.

Hint: use the double-angle formula first.

Working

$$\frac{du}{dx} = -\sin x \quad \Rightarrow \quad dx = -\frac{1}{\sin x}\,du.$$

$$\cos x = u - 1.$$

Using an inverse cosine for x would be more awkward.

$$\int \frac{2\sin 2x}{1+\cos x}\,dx = \int \frac{4\sin x\cos x}{1+\cos x}\,dx$$

$$= \int -\frac{4\sin x(u-1)}{u}\,\frac{1}{\sin x}\,du$$

$$= -4\int \frac{u-1}{u}\,du = -4\int \left(1 - \frac{1}{u}\right) du$$

$$= -4(u - \ln|u|) + C$$

$$= -4(1 + \cos x - \ln|1+\cos x|) + C$$

$$= 4\ln(1+\cos x) - 4\cos x + k.$$

Here $k = C - 4$, and $1 + \cos x > 0$ wherever the original expression is defined.

Definite Integration — Change the Limits

Solution 33

Calculate $\int_0^{\pi/2} \cos x\sqrt{1+\sin x}\,dx$.

Choose a substitution.

$u = 1 + \sin x.$

$\dfrac{du}{dx} = \cos x \quad \Rightarrow \quad du = \cos x\,dx.$

$\sin x = u - 1.$

Because we have changed from x to u, change the limits to values of u.

Find the new limits when $x = \pi/2$ and when $x = 0$.

$x = \pi/2 \Rightarrow u = 2, \qquad x = 0 \Rightarrow u = 1.$

$$\int_0^{\pi/2} \cos x\sqrt{1+\sin x}\,dx = \int_1^2 u^{1/2}\,du.$$

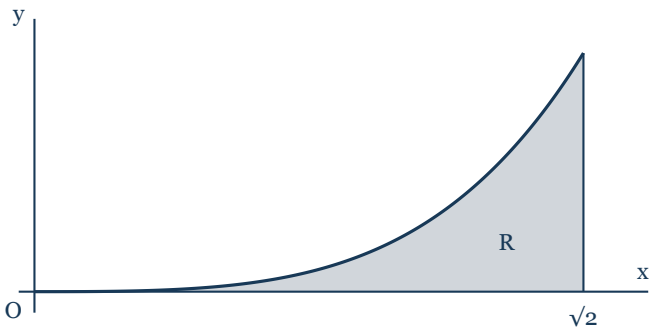
$$= \left[\frac{2}{3}u^{3/2}\right]_1^2 = \frac{2}{3}(2\sqrt{2}-1).$$

Test Your Understanding — Area Substitution

Solution 34

Edexcel C4 June 2011, Q4c

The curve is $y = x^3\ln(x^2+2)$, $x \geq 0$.



The finite region R is bounded by the curve, the x-axis and $x = \sqrt{2}$.

Use $u = x^2 + 2$ to show that its area is

$$\frac{1}{2}\int_2^4 (u-2)\ln u\,du.$$

Working

$\dfrac{du}{dx} = 2x \quad \Rightarrow \quad du = 2x\,dx.$

$x^2 = u - 2.$

When $x = \sqrt{2}$, $u = (\sqrt{2})^2 + 2 = 4.$

When $x = 0$, $u = 0^2 + 2 = 2.$

$$\int_0^{\sqrt{2}} x^2\ln(x^2+2)\,x\,dx$$

$$= \int_2^4 \frac{1}{2}(u-2)\ln u\,du.$$

5. Using $u^2 = 4x + 1$, or otherwise, find the exact value of $\int_6^{20} \frac{8x}{\sqrt{4x+1}}\,dx$.

5

Let $I = \int_6^{20} \frac{8x}{\sqrt{4x+1}}\,dx$, and take $u = \sqrt{4x+1}$.

$2u\frac{du}{dx} = 4 \quad \Rightarrow \quad dx = \frac{u}{2}\,du.$

The limits change: $x = 20 \Rightarrow u = 9$,

$x = 6 \Rightarrow u = 5$.

$$I = \int_5^9 \frac{2(u^2-1)}{u}\,\frac{u}{2}\,du = \left[\frac{u^3}{3} - u\right]_5^9$$

$$= \left(\frac{729}{3} - 9\right) - \left(\frac{125}{3} - 5\right) = \frac{592}{3}.$$

6. Use $u^2 = e^x - 2$ to show that

$$\int_{\ln 3}^{\ln 4} \frac{e^{4x}}{e^x - 2}\,dx = \frac{a}{b} + c\ln d,$$

where a, b, c, d are integers to be found.

6

Let the integral be I, and take $u = \sqrt{e^x - 2}$.

$2u\frac{du}{dx} = e^x = u^2 + 2 \quad \Rightarrow \quad dx = \frac{2u}{u^2+2}\,du.$

The limits change: $x = \ln 4 \Rightarrow u = \sqrt{2}$,

$x = \ln 3 \Rightarrow u = 1$.

$$I = \int_1^{\sqrt{2}} \frac{2(u^2+2)^3}{u}\,du$$

$$= \int_1^{\sqrt{2}} \left(2u^5 + 12u^3 + 24u + \frac{16}{u}\right)du$$

$$= \left[\frac{1}{3}u^6 + 3u^4 + 12u^2 + 16\ln|u|\right]_1^{\sqrt{2}}$$

$$= \frac{70}{3} + 8\ln 2.$$

$a = 70, \quad b = 3, \quad c = 8, \quad d = 2.$

7. Prove that $-\int \frac{1}{\sqrt{1-x^2}}\,dx = \arccos x + C$.

7

Let $x = \cos\theta$, with $0 < \theta < \pi$, so $-1 < x < 1$.

$\frac{dx}{d\theta} = -\sin\theta \quad \Rightarrow \quad dx = -\sin\theta\,d\theta.$

$\sqrt{1-x^2} = \sqrt{1-\cos^2\theta} = \sin\theta.$

$$I = -\int \frac{1}{\sin\theta}(-\sin\theta)\,d\theta = \int 1\,d\theta$$

$$= \theta + C = \arccos x + C.$$

8. Use $u = \cos x$ to show that

$$\int_0^{\pi/3} \sin^3 x\cos^2 x\,dx = \frac{47}{480}.$$

8

Let the integral be I.

$u = \cos x, \quad \frac{du}{dx} = -\sin x, \quad dx = -\frac{du}{\sin x}.$

Limits: $x = \pi/3 \Rightarrow u = 1/2$; $x = 0 \Rightarrow u = 1$.

$$I = \int_1^{1/2} -(1-u^2)u^2\,du$$

$$= \int_1^{1/2} (u^2-1)u^2\,du = \int_1^{1/2} (u^4 - u^2)\,du$$

$$= \left[\frac{u^5}{5} - \frac{u^3}{3}\right]_1^{1/2}$$

$$= \left(\frac{1}{160} - \frac{1}{24}\right) - \left(\frac{1}{5} - \frac{1}{3}\right) = \frac{47}{480}.$$

Exercise 11E — Page 307 Solution 37

9. Using a suitable trigonometric substitution for x, find

$$\int_{1/2}^{\sqrt{3}/2} x^2\sqrt{1-x^2}\,dx.$$

Challenge

Using a substitution of the form $x = k\sin u$, show that

$$\int \frac{1}{x^2\sqrt{9-x^2}}\,dx = -\frac{\sqrt{9-x^2}}{9x} + C.$$

9

Let the integral be I, and put $x = \sin\theta$. Then $dx = \cos\theta\,d\theta$.

Limits: $x = \sqrt{3}/2 \Rightarrow \theta = \pi/3$, $x = 1/2 \Rightarrow \theta = \pi/6$.

$$I = \int_{\pi/6}^{\pi/3} \sin^2\theta\cos^2\theta\,d\theta$$

$$= \int_{\pi/6}^{\pi/3} \frac{1}{4}\sin^2 2\theta\,d\theta = \int_{\pi/6}^{\pi/3} \frac{1}{8}(1-\cos 4\theta)\,d\theta$$

$$= \left[\frac{1}{8}\left(\theta - \frac{1}{4}\sin 4\theta\right)\right]_{\pi/6}^{\pi/3}$$

$$= \frac{1}{8}\left[\left(\frac{\pi}{3} + \frac{\sqrt{3}}{8}\right) - \left(\frac{\pi}{6} - \frac{\sqrt{3}}{8}\right)\right]$$

$$= \frac{1}{8}\left(\frac{\pi}{6} + \frac{\sqrt{3}}{4}\right) = \frac{2\pi + 3\sqrt{3}}{96}.$$

Challenge

Let $x = 3\sin u$, with $-\pi/2 < u < \pi/2$, excluding $u = 0$.

$$\frac{dx}{du} = 3\cos u \quad \Rightarrow \quad dx = 3\cos u\,du.$$

$$I = \int \frac{3\cos u}{9\sin^2 u\sqrt{9-9\sin^2 u}}\,du$$

$$= \int \frac{3\cos u}{9\sin^2 u\,3\cos u}\,du = \frac{1}{9}\int \operatorname{cosec}^2 u\,du$$

$$= -\frac{1}{9}\cot u + C = -\frac{\cos u}{9\sin u} + C.$$

$$\cos u = \sqrt{1-\sin^2 u} = \sqrt{1-\frac{x^2}{9}} = \frac{\sqrt{9-x^2}}{3}.$$

$$I = -\frac{\sqrt{9-x^2}/3}{9x/3} + C = -\frac{\sqrt{9-x^2}}{9x} + C.$$

Integration by Parts Solution 38

How can we find $\int x\cos x\,dx$?

Just as the product rule differentiates a product of two expressions, integration by parts can often integrate a product.

Integration by parts

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx.$$

Proof (not needed for the examination)

The product rule gives

$$\frac{d}{dx}(uv) = v\frac{du}{dx} + u\frac{dv}{dx}.$$

Both terms on the right are products. We can let $u\frac{dv}{dx}$ represent a product such as $x\cos x$.

Rearranging:

$$u\frac{dv}{dx} = \frac{d}{dx}(uv) - v\frac{du}{dx}.$$

Integrating both sides with respect to x gives the formula above.

Integration by Parts — Choosing the Factors

Solution 39

Find $\int x \cos x \, dx$.

Step 1

Decide which factor is u and which is dv/dx.

Choose u so that differentiating makes it simpler, and choose dv/dx so that it is easy to integrate.

For the standard products here, choose the x^n term as u, unless the other factor is $\ln x$.

$$u = x, \qquad \frac{dv}{dx} = \cos x.$$

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx.$$

Step 2

Find du/dx and v.

$$\frac{du}{dx} = 1, \qquad v = \sin x.$$

Step 3

Use the formula.

$$\int x \cos x \, dx = x \sin x - \int 1 \sin x \, dx$$
$$= x \sin x + \cos x + C.$$

Memory aid: "uv minus the integral of the two new expressions multiplied together".

Another Integration-by-Parts Example

Solution 40

Find $\int x^2 \ln x \, dx$, for $x > 0$.

Step 1

Choose u and dv/dx.

Step 2

Find du/dx and v.

Step 3

Use the integration-by-parts formula.

Working

Here choose $\ln x$ as u, because its derivative is simpler.

$$u = \ln x, \qquad \frac{dv}{dx} = x^2.$$
$$\frac{du}{dx} = \frac{1}{x}, \qquad v = \frac{1}{3}x^3.$$
$$\int x^2 \ln x \, dx = \frac{1}{3}x^3 \ln x - \int \frac{1}{3}x^2 \, dx$$
$$= \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C.$$

Integration by Parts Twice

Solution 41

Find $\int x^2 e^x \, dx$.

Tip: write the working for the second integral separately, then substitute its result into the original calculation.

Working

$u = x^2, \quad \dfrac{dv}{dx} = e^x, \quad \dfrac{du}{dx} = 2x, \quad v = e^x.$

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx.$$

Apply integration by parts again:

$u = 2x, \quad \dfrac{dv}{dx} = e^x, \quad \dfrac{du}{dx} = 2, \quad v = e^x.$

$$\int 2x e^x \, dx = 2x e^x - \int 2 e^x \, dx = 2x e^x - 2e^x + C_1.$$

Therefore

$$\int x^2 e^x \, dx = x^2 e^x - (2x e^x - 2e^x) + C$$

$$= x^2 e^x - 2x e^x + 2e^x + C.$$

Test Your Understanding — Parts Twice

Solution 42

Find $\int x^2 \sin x \, dx$.

Answer

$$2x \sin x - x^2 \cos x + 2 \cos x + C.$$

Integrating Logarithms and Definite Integrals — Solution 43

Find $\int \ln x \, dx$, for $x > 0$.

Indefinite

$u = \ln x, \qquad \dfrac{dv}{dx} = 1.$

$\dfrac{du}{dx} = \dfrac{1}{x}, \qquad v = x.$

$$\int \ln x \, dx = x \ln x - \int 1 \, dx$$

$$= x \ln x - x + C.$$

General definite-integral formula

$$\int_a^b u \frac{dv}{dx} \, dx = [uv]_a^b - \int_a^b v \frac{du}{dx} \, dx.$$

Find $\int_1^2 \ln x \, dx$, leaving the answer in terms of natural logarithms.

Definite

$$\int_1^2 \ln x \, dx = [x \ln x - x]_1^2 = 2 \ln 2 - 1.$$

Alternatively, from scratch:

$u = \ln x, \quad \dfrac{dv}{dx} = 1, \quad \dfrac{du}{dx} = \dfrac{1}{x}, \quad v = x.$

$$\int_1^2 \ln x \, dx = [x \ln x]_1^2 - \int_1^2 1 \, dx$$

$$= 2 \ln 2 - 1 \ln 1 - [x]_1^2$$

$$= 2 \ln 2 - (2 - 1) = 2 \ln 2 - 1.$$

Test Your Understanding — Definite Integration by Parts — Solution 44

Find $\int_0^{\pi/2} x \sin x \, dx$.

Working

$u = x, \quad \dfrac{dv}{dx} = \sin x, \quad \dfrac{du}{dx} = 1, \quad v = -\cos x.$

$$\int x \sin x \, dx = -x \cos x - \int (-\cos x) \, dx$$

$$= -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + C.$$

$$\therefore \int_0^{\pi/2} x \sin x \, dx = [-x \cos x + \sin x]_0^{\pi/2}$$

$$= \left(-\frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2}\right) - (0 + \sin 0) = 1.$$

When the Original Integral Returns

Solution 45

Find $\int e^x \sin x \, dx$.

Either factor can be chosen as u. The key is to recognise the original integral when it reappears.

When the same integral returns, collect those terms together.

Working

Let $I = \int e^x \sin x \, dx$.

$u = e^x, \quad \frac{dv}{dx} = \sin x, \quad \frac{du}{dx} = e^x, \quad v = -\cos x.$

$I = -e^x \cos x - \int (-e^x \cos x) \, dx$

$= -e^x \cos x + \int e^x \cos x \, dx.$

Now consider $\int e^x \cos x \, dx$:

$u = e^x, \quad \frac{dv}{dx} = \cos x, \quad \frac{du}{dx} = e^x, \quad v = \sin x.$

$I = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx.$

$\Longrightarrow$ Collect the two copies of I:

$2I = -e^x \cos x + e^x \sin x + C.$

$$\int e^x \sin x \, dx = -\frac{1}{2}e^x \cos x + \frac{1}{2}e^x \sin x + C'.$$

Exercise 11F — Integration by Parts

Solution 46

5a. Use integration by parts to find $\int x \cos 4x \, dx$.

5a

Let $I = \int x \cos 4x \, dx$.

$u = x \Rightarrow \frac{du}{dx} = 1.$

$\frac{dv}{dx} = \cos 4x \Rightarrow v = \frac{1}{4}\sin 4x.$

$I = \frac{x}{4}\sin 4x - \int \frac{1}{4}\sin 4x \, dx$

$= \frac{x}{4}\sin 4x + \frac{1}{16}\cos 4x + C$

$= \frac{1}{16}(4x \sin 4x + \cos 4x) + C.$

5b. Use your answer to part a to find $\int x^2 \sin 4x \, dx$.

5b

Let $I = \int x^2 \sin 4x \, dx$.

$u = x^2 \Rightarrow \frac{du}{dx} = 2x.$

$\frac{dv}{dx} = \sin 4x \Rightarrow v = -\frac{1}{4}\cos 4x.$

$I = -\frac{x^2}{4}\cos 4x + \int \frac{x}{2}\cos 4x \, dx$

$= -\frac{x^2}{4}\cos 4x + \frac{1}{32}(4x \sin 4x + \cos 4x) + C$

$= \frac{1}{32}[(1 - 8x^2)\cos 4x + 4x \sin 4x] + C.$

Exercise 11F — Integration by Parts

Solution 47

6a. Find $\int \sqrt{8-x}\,dx$.

b. Using integration by parts or otherwise, show that

$$\int (x-2)\sqrt{8-x}\,dx = -\frac{2}{5}(8-x)^{3/2}(x+2) + C.$$

c. Hence find $\int_4^7 (x-2)\sqrt{8-x}\,dx$.

6a

$$\int \sqrt{8-x}\,dx = \int (8-x)^{1/2}\,dx$$
$$= -\frac{2}{3}(8-x)^{3/2} + C.$$

6c

$$\int_4^7 (x-2)\sqrt{8-x}\,dx$$
$$= \left[-\frac{2}{5}(8-x)^{3/2}(x+2)\right]_4^7$$
$$= -\frac{2}{5}\times 9 + \frac{2}{5}\times 48$$
$$= \frac{78}{5} = 15.6.$$

6b

Let $I = \int (x-2)\sqrt{8-x}\,dx$.

$u = x-2, \quad \frac{du}{dx} = 1.$

$\frac{dv}{dx} = \sqrt{8-x}, \quad v = -\frac{2}{3}(8-x)^{3/2}.$

$$I = -\frac{2}{3}(x-2)(8-x)^{3/2} + \frac{2}{3}\int (8-x)^{3/2}\,dx$$
$$= -\frac{2}{3}(x-2)(8-x)^{3/2} - \frac{4}{15}(8-x)^{5/2} + C$$
$$= -\frac{2}{3}(x-2)(8-x)^{3/2} - \frac{4}{15}(8-x)^{3/2}(8-x) + C$$
$$= \frac{2}{3}(2-x)(8-x)^{3/2} + \frac{4}{15}(8-x)^{3/2}(x-8) + C$$
$$= \frac{2}{15}(8-x)^{3/2}[5(2-x) + 2(x-8)] + C$$
$$= \frac{2}{15}(8-x)^{3/2}(-3x-6) + C$$
$$= -\frac{2}{5}(8-x)^{3/2}(x+2) + C.$$

Exercise 11F — Integration by Parts

Solution 48

7a. Find $\int \sec^2 3x\,dx$.

b. Using integration by parts or otherwise, find $\int x\sec^2 3x\,dx$.

c. Hence show that $\int_{\pi/18}^{\pi/9} x\sec^2 3x\,dx = p\pi - q\ln 3$, finding the exact values of p and q.

7a

$$\int \sec^2 3x\,dx = \frac{1}{3}\tan 3x + C.$$

7b

Let $I = \int x\sec^2 3x\,dx$.

$u = x \Rightarrow \frac{du}{dx} = 1.$

$\frac{dv}{dx} = \sec^2 3x \Rightarrow v = \frac{1}{3}\tan 3x.$

$$I = \frac{x}{3}\tan 3x - \frac{1}{3}\int \tan 3x\,dx$$
$$= \frac{x}{3}\tan 3x - \frac{1}{9}\ln|\sec 3x| + C.$$

7c

$$\int_{\pi/18}^{\pi/9} x\sec^2 3x\,dx = \left[\frac{x}{3}\tan 3x - \frac{1}{9}\ln|\sec 3x|\right]_{\pi/18}^{\pi/9}$$
$$= \left(\frac{\pi}{27}\sqrt{3} - \frac{1}{9}\ln 2\right) - \left(\frac{\pi}{54\sqrt{3}} - \frac{1}{9}\ln\frac{2}{\sqrt{3}}\right)$$
$$= \left(\frac{\sqrt{3}\pi}{27} - \frac{1}{9}\ln 2\right) - \left(\frac{\sqrt{3}\pi}{162} - \frac{1}{9}\ln\frac{2}{\sqrt{3}}\right)$$
$$= \frac{5\sqrt{3}\pi}{162} - \frac{1}{9}\ln 2 + \frac{1}{9}\ln 2 - \frac{1}{9}\ln\sqrt{3}$$
$$= \frac{5\sqrt{3}\pi}{162} - \frac{1}{18}\ln 3.$$

$p = \frac{5\sqrt{3}}{162}, \qquad q = \frac{1}{18}.$

Exercise 11F — Extensions and Standard Results

Solution 49

Extension

Find $\int \arcsin x \, dx$.

Hint: choose $dv/dx = 1$.

Extension

$\sqrt{1-x^2} + x \arcsin x + C.$

STEP I 2014, Q2

i. Show that

$$\int \ln(2-x)\, dx$$
$$= -(2-x)\ln(2-x) + (2-x) + C$$

where $x < 2$.

ii. Sketch curve A, given by $y = \ln|x^2 - 4|$.

iii. Show that the area of the finite region enclosed by the positive x-axis, the y-axis and curve A is $4\ln(2+\sqrt{3}) - 2\sqrt{3}$.

iv. Curve B is given by $y = |\ln|x^2 - 4||$. Find the area between B and the x-axis with $|x| < 2$.

Note: you may assume that $t \ln t \to 0$ as $t \to 0^+$.

You will need these standard results from the formula booklet. We will prove them later.

$$\int \tan x \, dx = \ln|\sec x| + C.$$

$$\int \cot x \, dx = \ln|\sin x| + C.$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C.$$

$$\int \operatorname{cosec} x \, dx = -\ln|\operatorname{cosec} x + \cot x| + C.$$

Using Partial Fractions

Solution 50

Partial fractions split some rational expressions into simpler terms. This lets us integrate expressions with more complicated denominators.

Find $\int \frac{2}{x^2-1}\, dx$.

Working

$$\frac{2}{x^2-1} = \frac{2}{(x+1)(x-1)}$$
$$= \frac{A}{x+1} + \frac{B}{x-1}.$$
$$2 = A(x-1) + B(x+1) = Ax - A + Bx + B.$$
$$A + B = 0, \qquad B - A = 2 \quad \Rightarrow \quad B = 1, \quad A = -1.$$
$$\int \frac{2}{x^2-1}\, dx = \int \left(-\frac{1}{x+1} + \frac{1}{x-1}\right) dx$$
$$= -\ln|x+1| + \ln|x-1| + C$$
$$= \ln\left|\frac{x-1}{x+1}\right| + C.$$

Further Partial-Fraction Examples

Solution 51

Find $\int \frac{x-5}{(x+1)(x-2)}\,dx$.

$$\frac{x-5}{(x+1)(x-2)} \equiv \frac{A}{x+1} + \frac{B}{x-2}.$$

$$A = 2, \qquad B = -1.$$

$$\int \frac{x-5}{(x+1)(x-2)}\,dx = \int \left(\frac{2}{x+1} - \frac{1}{x-2}\right) dx$$

$$= 2\ln|x+1| - \ln|x-2| + C$$

$$= \ln\left|\frac{(x+1)^2}{x-2}\right| + C.$$

Find $\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2}\,dx$.

$$\frac{8x^2-19x+1}{(2x+1)(x-2)^2}$$

$$= \frac{A}{2x+1} + \frac{B}{(x-2)^2} + \frac{D}{x-2}.$$

$$A = 2, \quad B = -1, \quad D = 3.$$

$$\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2}\,dx$$

$$= \ln|(2x+1)(x-2)^3| + \frac{1}{x-2} + C.$$

Test Your Understanding — Partial Fractions

Solution 52

Edexcel C4 June 2009, Q3

$$f(x) = \frac{4-2x}{(2x+1)(x+1)(x+3)} = \frac{A}{2x+1} + \frac{B}{x+1} + \frac{C}{x+3}.$$

a. Find A, B, C.

b(i). Hence find $\int f(x)\,dx$.

b(ii) Find $\int_0^2 f(x)\,dx$ in the form $\ln k$.

a

$$4-2x = A(x+1)(x+3) + B(2x+1)(x+3) + C(2x+1)(x+1).$$

$x = -\frac{1}{2}$: $\quad 5 = A(\frac{1}{2})(\frac{5}{2}) \Rightarrow A = 4.$

$x = -1$: $\quad 6 = B(-1)(2) \Rightarrow B = -3.$

$x = -3$: $\quad 10 = C(-5)(-2) \Rightarrow C = 1.$

bi

$$\int f(x)\,dx = \int \left(\frac{4}{2x+1} - \frac{3}{x+1} + \frac{1}{x+3}\right) dx$$

$$= \frac{4}{2}\ln|2x+1| - 3\ln|x+1| + \ln|x+3| + K.$$

bii

$$\int_0^2 f(x)\,dx$$

$$= [2\ln(2x+1) - 3\ln(x+1) + \ln(x+3)]_0^2$$

$$= (2\ln 5 - 3\ln 3 + \ln 5) - (2\ln 1 - 3\ln 1 + \ln 3)$$

$$= 3\ln 5 - 4\ln 3 = \ln\left(\frac{5^3}{3^4}\right)$$

$$= \ln\frac{125}{81}.$$

Integrating Improper Algebraic Fractions

Solution 53

Find $\int \frac{x^2}{x+1}\,dx$.

How can we simplify a fraction whose numerator has the higher degree?

Simplify

Polynomial division:

$$\begin{array}{r|l} & x \quad -1 \\ \hline x+1 & x^2 +0x +0 \\ & x^2 \quad +x \\ \hline & \quad -x +0 \\ & \quad -x -1 \\ \hline & \qquad 1 \end{array}$$

$$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}.$$

Now integrate.

Integrate

$$\int \frac{x^2}{x+1}\,dx = \int \left(x - 1 + \frac{1}{x+1}\right) dx$$

$$= \frac{1}{2}x^2 - x + \ln|x+1| + C.$$

Test Your Understanding — Algebraic Fractions

Solution 54

Find $\int \frac{x}{x-1}\,dx$.

$$\int \frac{x}{x-1}\,dx = \int \frac{x-1+1}{x-1}\,dx$$

$$= \int \left(1 + \frac{1}{x-1}\right) dx = x + \ln|x-1| + C.$$

$\Longrightarrow$ Compare this with $\int \frac{x-1}{x}\,dx$, which is simpler:

$$\int \frac{x-1}{x}\,dx = \int \left(1 - \frac{1}{x}\right) dx = x - \ln|x| + C.$$

Find $\int \frac{x^3+2}{x+1}\,dx$.

$$\frac{1}{3}x^3 - \frac{1}{2}x^2 + x + \ln|x+1| + C.$$

Exercise 11G — Pages 312–313 Solution 55

3. $f(x) = \frac{4}{(2x+1)(1-2x)}$, $x \neq \pm\frac{1}{2}$.

a. Given $f(x) = \frac{A}{2x+1} + \frac{B}{1-2x}$, find A and B.

b. Hence find $\int f(x)\,dx$, as a single logarithm plus a constant.

c. Find $\int_1^2 f(x)\,dx$ in the form $\ln k$, with k rational.

3a

$$\frac{4}{(2x+1)(1-2x)} = \frac{A}{2x+1} + \frac{B}{1-2x}.$$

$$4 = A(1-2x) + B(2x+1).$$

$x = \frac{1}{2}$: $\quad 4 = 2B \Rightarrow B = 2.$

$x = -\frac{1}{2}$: $\quad 4 = 2A \Rightarrow A = 2.$

3b

$$\int f(x)\,dx = \int \left(\frac{2}{2x+1} + \frac{2}{1-2x}\right) dx$$

$$= \ln|2x+1| - \ln|1-2x| + C$$

$$= \ln\left|\frac{2x+1}{1-2x}\right| + C.$$

3c

$$\int_1^2 f(x)\,dx = \left[\ln\left|\frac{2x+1}{1-2x}\right|\right]_1^2$$

$$= \ln\frac{5}{3} - \ln 3 = \ln\frac{5}{9}.$$

$$k = \frac{5}{9}.$$

Exercise 11G — Pages 312–313 Solution 56

4. $f(x) = \frac{17-5x}{(3+2x)(2-x)^2}$, $-\frac{3}{2} < x < 2$.

a. Express $f(x)$ in partial fractions.

b. Hence find $\int_0^1 f(x)\,dx$ in the form $a + \ln b$.

4a

$$\frac{17-5x}{(3+2x)(2-x)^2}$$

$$= \frac{A}{3+2x} + \frac{B}{(2-x)^2} + \frac{C}{2-x}.$$

$$17 - 5x = A(2-x)^2 + B(3+2x) + C(3+2x)(2-x).$$

$x = 2$: $\quad 7 = 7B \Rightarrow B = 1.$

$x = -\frac{3}{2}$: $\quad 17 + \frac{15}{2} = \frac{49}{4}A \Rightarrow A = 2.$

$x = 0$: $\quad 17 = 4A + 3B + 6C = 8 + 3 + 6C.$

$$C = 1.$$

$$f(x) = \frac{2}{3+2x} + \frac{1}{(2-x)^2} + \frac{1}{2-x}.$$

4b

$$\int_0^1 \left(\frac{2}{3+2x} + \frac{1}{2-x} + \frac{1}{(2-x)^2}\right) dx$$

$$= \left[\ln|3+2x| - \ln|2-x| + \frac{1}{2-x}\right]_0^1$$

$$= (\ln 5 - \ln 1 + 1) - (\ln 3 - \ln 2 + \tfrac{1}{2})$$

$$= \frac{1}{2} + \ln\frac{10}{3}.$$

$$a = \frac{1}{2}, \qquad b = \frac{10}{3}.$$

Exercise 11G — Pages 312–313 Solution 57

5. $f(x) = \frac{9x^2+4}{9x^2-4}$, $x \neq \pm\frac{2}{3}$.

a. Given $f(x) = A + \frac{B}{3x-2} + \frac{C}{3x+2}$, find A, B, C.

b. Hence find $\int_{-1/3}^{1/3} f(x)\,dx$ in the form $a + b\ln c$, with a, b, c rational.

5a

Dividing gives

$$f(x) = 1 + \frac{8}{9x^2-4} = 1 + \frac{8}{(3x+2)(3x-2)}.$$

$$\frac{8}{(3x+2)(3x-2)} = \frac{B}{3x-2} + \frac{C}{3x+2}.$$

$$8 = B(3x+2) + C(3x-2).$$

$x = -\frac{2}{3}$: $\quad 8 = -4C \Rightarrow C = -2.$

$x = \frac{2}{3}$: $\quad 8 = 4B \Rightarrow B = 2.$

$$A = 1, \quad B = 2, \quad C = -2.$$

5b

$$\int_{-1/3}^{1/3} \left(1 + \frac{2}{3x-2} - \frac{2}{3x+2}\right) dx$$

$$= \left[x + \frac{2}{3}\ln|3x-2| - \frac{2}{3}\ln|3x+2|\right]_{-1/3}^{1/3}$$

$$= \left(\frac{1}{3} - \frac{2}{3}\ln 3\right) - \left(-\frac{1}{3} + \frac{2}{3}\ln 3\right)$$

$$= \frac{2}{3} - \frac{4}{3}\ln 3.$$

$$a = \frac{2}{3}, \quad b = -\frac{4}{3}, \quad c = 3.$$

Exercise 11G — Pages 312–313 Solution 58

6. $f(x) = \frac{6+3x-x^2}{x^3+2x^2}$, $x > 0$.

a. Express $f(x)$ in partial fractions.

b. Hence find $\int_2^4 f(x)\,dx$ in the form $a + \ln b$, with a, b rational.

6a

$$f(x) = \frac{6+3x-x^2}{x^2(x+2)} = \frac{A}{x^2} + \frac{B}{x} + \frac{C}{x+2}.$$

$$6 + 3x - x^2 = A(x+2) + Bx(x+2) + Cx^2.$$

$x = 0$: $\quad 6 = 2A \Rightarrow A = 3.$

$x = -2$: $\quad -4 = 4C \Rightarrow C = -1.$

$x = 1$: $\quad 8 = 3A + 3B + C \Rightarrow B = 0.$

$$f(x) = \frac{3}{x^2} - \frac{1}{x+2}.$$

6b

$$\int_2^4 f(x)\,dx = \int_2^4 \left(\frac{3}{x^2} - \frac{1}{x+2}\right) dx$$

$$= \left[-\frac{3}{x} - \ln|x+2|\right]_2^4$$

$$= \left(-\frac{3}{4} - \ln 6\right) - \left(-\frac{3}{2} - \ln 4\right)$$

$$= \frac{3}{4} + \ln\frac{2}{3}.$$

$$a = \frac{3}{4}, \qquad b = \frac{2}{3}.$$

Exercise 11G — Pages 312–313 Solution 59

7. $\frac{32x^2+4}{(4x+1)(4x-1)} = A + \frac{B}{4x+1} + \frac{C}{4x-1}$.

a. Find A, B, C.

b. Hence find $\int_1^2 \frac{32x^2+4}{(4x+1)(4x-1)}\,dx$ in the form $2 + k\ln m$, with k, m rational.

7a

Let $f(x) = \frac{32x^2+4}{(4x+1)(4x-1)}$. Dividing:

$$f(x) = 2 + \frac{6}{(4x+1)(4x-1)} \quad \Rightarrow \quad A = 2.$$

$$\frac{6}{(4x+1)(4x-1)} = \frac{B}{4x+1} + \frac{C}{4x-1}.$$

$$6 = B(4x-1) + C(4x+1).$$

$x = \frac{1}{4}$: $\quad 6 = 2C \Rightarrow C = 3.$

$x = -\frac{1}{4}$: $\quad 6 = -2B \Rightarrow B = -3.$

$$f(x) = 2 - \frac{3}{4x+1} + \frac{3}{4x-1}.$$

7b

$$\int_1^2 f(x)\,dx = \int_1^2 \left(2 - \frac{3}{4x+1} + \frac{3}{4x-1}\right) dx$$

$$= \left[2x - \frac{3}{4}\ln|4x+1| + \frac{3}{4}\ln|4x-1|\right]_1^2$$

$$= \left(4 - \frac{3}{4}\ln 9 + \frac{3}{4}\ln 7\right) - \left(2 - \frac{3}{4}\ln 5 + \frac{3}{4}\ln 3\right)$$

$$= 2 + \frac{3}{4}(-\ln 9 + \ln 7 + \ln 5 - \ln 3)$$

$$= 2 + \frac{3}{4}\ln\frac{35}{27}.$$

$$k = \frac{3}{4}, \qquad m = \frac{35}{27}.$$

Exercise 11G — Extension Solution 60

Extension — MAT 2003, 1D

What is the exact value of

$$\int_1^2 \frac{dx}{x + x^3}?$$

Solution

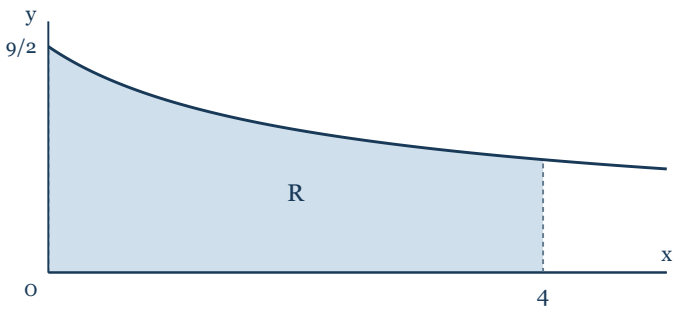
$$\frac{1}{2}\ln\frac{8}{5}.$$

Finding Areas

Solution 61

Definite integration gives signed area between a curve and the x-axis; when the curve is above the axis, this is its area. Our new integration skills allow more types of curve.

The curve is $y = 9/\sqrt{4+3x}$. Region R is bounded by the curve, the x-axis and $x = 0$, $x = 4$. Find its area.



Area

$$\text{Area} = \int_0^4 \frac{9}{\sqrt{4+3x}}\,dx$$

$$= 9\int_0^4 (4+3x)^{-1/2}\,dx$$

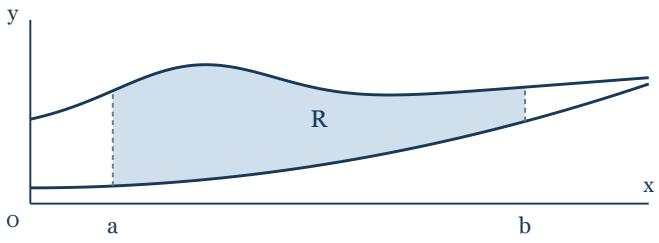
$$= 6\left[(4+3x)^{1/2}\right]_0^4 = 6(4-2) = 12.$$

Recall: when integrating $(a+bx)^n$, divide by the new power $n+1$ and by b.

Area Between Two Curves

Solution 62

Top: y = f(x); bottom: y = g(x)



For $f(x) \geq g(x)$ throughout $[a,b]$, the area between the curves is

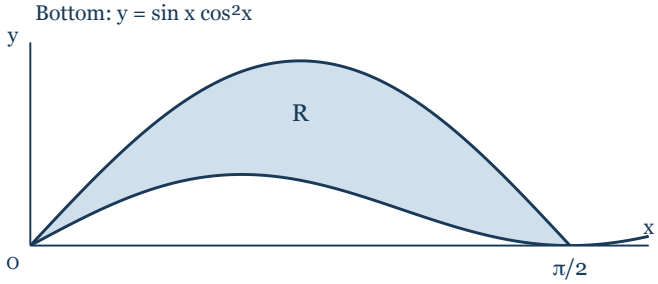
$$R = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

$$= \int_a^b [f(x) - g(x)]\,dx.$$

Use top curve minus bottom curve. Split the interval if their order changes.

The curves are $y = \sin 2x$ and $y = \sin x \cos^2 x$, for $0 \leq x \leq \pi/2$. Find the area R bounded by them.

Top: y = sin 2x
Bottom: y = sin x cos²x



Area

$$\text{Area} = \int_0^{\pi/2} (\sin 2x - \sin x \cos^2 x)\,dx$$

$$= \left[-\frac{1}{2}\cos 2x + \frac{1}{3}\cos^3 x\right]_0^{\pi/2}$$

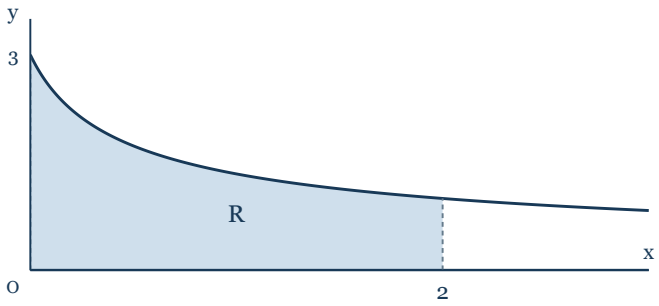
$$= \frac{1}{2} - \left(-\frac{1}{2} + \frac{1}{3}\right) = \frac{2}{3}.$$

Test Your Understanding — Finding Areas

Solution 63

Edexcel C4 January 2009, Q2a

The curve is $y = 3/\sqrt{1+4x}$. Region R is bounded by the curve, the x-axis and $x = 0$, $x = 2$. Use integration to find its area.



Area

$$\text{Area}(R) = \int_0^2 \frac{3}{\sqrt{1+4x}}\,dx = \int_0^2 3(1+4x)^{-1/2}\,dx$$

$$= \left[\frac{3(1+4x)^{1/2}}{\frac{1}{2}\times 4}\right]_0^2$$

$$= \left[\frac{3}{2}(1+4x)^{1/2}\right]_0^2$$

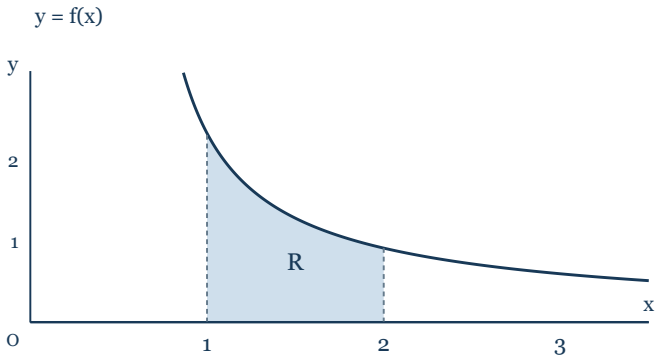
$$= \frac{3}{2}\sqrt{9} - \frac{3}{2}(1) = \frac{9}{2} - \frac{3}{2} = 3 \text{ square units.}$$

Exercise 11H — Pages 314–317

Solution 64

3. $f(x) = \frac{4x+3}{(x+2)(2x-1)}$, $x > \frac{1}{2}$.

Find the shaded area bounded by $y = f(x)$, the x-axis and $x = 1$, $x = 2$.



3

$$\text{Area} = \int_1^2 \frac{4x+3}{(x+2)(2x-1)}\,dx.$$

$$\frac{4x+3}{(x+2)(2x-1)} = \frac{A}{x+2} + \frac{B}{2x-1}.$$

$$4x+3 = A(2x-1) + B(x+2).$$

$$x = -2: \quad -5 = -5A \Rightarrow A = 1.$$

$$x = \tfrac{1}{2}: \quad 5 = \tfrac{5}{2}B \Rightarrow B = 2.$$

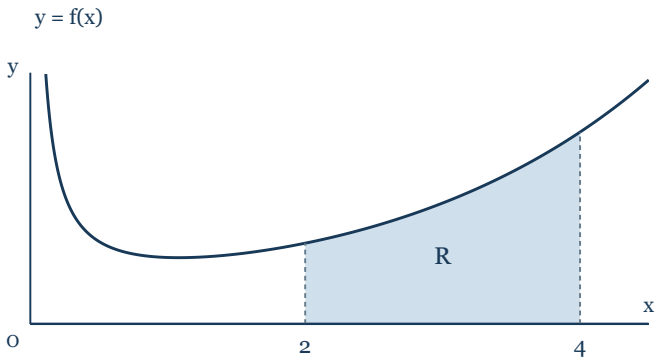
$$\text{Area} = \int_1^2 \left(\frac{1}{x+2} + \frac{2}{2x-1}\right)dx$$

$$= \left[\ln|x+2| + \ln|2x-1|\right]_1^2$$

$$= (\ln 4 + \ln 3) - (\ln 3 + \ln 1) = \ln 4.$$

4. $f(x) = e^{0.5x} + 1/x,\ x > 0.$

Find the shaded area bounded by $y = f(x)$, the x-axis and $x = 2$, $x = 4$.



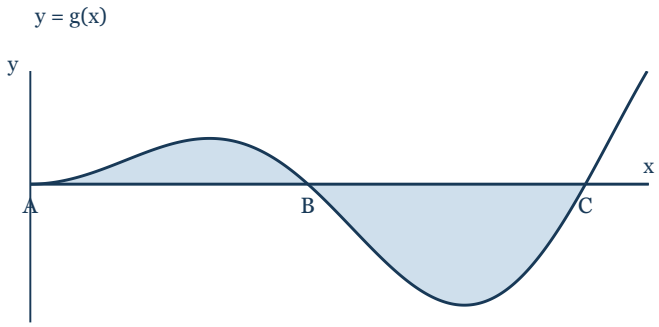
4

$$\text{Area} = \int_2^4 \left(e^{0.5x} + \frac{1}{x}\right) dx$$
$$= [2e^{0.5x} + \ln|x|]_2^4$$
$$= (2e^2 + \ln 4) - (2e + \ln 2)$$
$$= 2e^2 - 2e + \ln\frac{4}{2}$$
$$= 2e^2 - 2e + \ln 2.$$

5. The curve is $y = g(x) = x\sin x$.

a. Write down the coordinates of A, B, C.

b. Find the total shaded area.



5a

$g(x) = 0 \Rightarrow A(0,0),\ B(\pi,0),\ C(2\pi,0).$

5b

$$I = \int x\sin x\,dx.$$

$$u = x \Rightarrow \frac{du}{dx} = 1; \qquad \frac{dv}{dx} = \sin x \Rightarrow v = -\cos x.$$

$$I = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x + C.$$

Between A and B:

$$[-x\cos x + \sin x]_0^{\pi} = \pi.$$

Between B and C, the signed integral is

$$[-x\cos x + \sin x]_{\pi}^{2\pi} = -2\pi - \pi = -3\pi.$$

The area there is 3π.

Total area $= \pi + 3\pi = 4\pi$.

Exercise 11H — Pages 314–317 — Solution 67

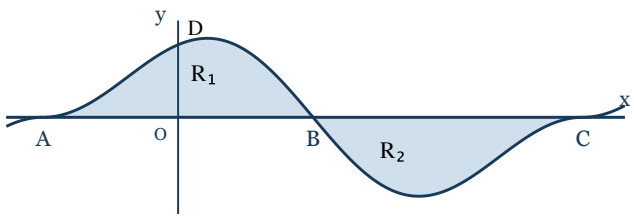
7. The curve is $y = 3\cos x\sqrt{\sin x + 1}$.

a. Find A, B, C, D.

b. Use a suitable substitution to find $\int 3\cos x\sqrt{\sin x + 1}\,dx$.

c. Show the two shaded regions have the same area and express that area as $\sqrt{a}$, finding the positive integer a.

D is the y-intercept; shaded regions R_1 and R_2



7a

On the x-axis, $y = 0$, so $\cos x = 0$ or $\sin x = -1$.

$x = -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$.

$A(-\frac{\pi}{2}, 0),\ B(\frac{\pi}{2}, 0),\ C(\frac{3\pi}{2}, 0)$.

At $x = 0$, $y = 3$, so $D(0, 3)$.

7b

Let $u = \sin x + 1$, so $du/dx = \cos x$.

$$I = \int 3\sqrt{u}\,du = 2u^{3/2} + C$$

$$= 2(\sin x + 1)^{3/2} + C.$$

7c

$$R_1 = \int_{-\pi/2}^{\pi/2} 3\cos x\sqrt{\sin x + 1}\,dx$$

$$= [2(\sin x + 1)^{3/2}]_{-\pi/2}^{\pi/2} = 4\sqrt{2} = \sqrt{32}.$$

The second region lies below the axis:

$$\int_{\pi/2}^{3\pi/2} 3\cos x\sqrt{\sin x + 1}\,dx$$

$$= [2(\sin x + 1)^{3/2}]_{\pi/2}^{3\pi/2} = -4\sqrt{2}.$$

Its area is $R_2 = 4\sqrt{2}$. Therefore

$R_1 = R_2 = \sqrt{32}$, $\quad a = 32$.

Exercise 11H — Pages 314–317 — Solution 68

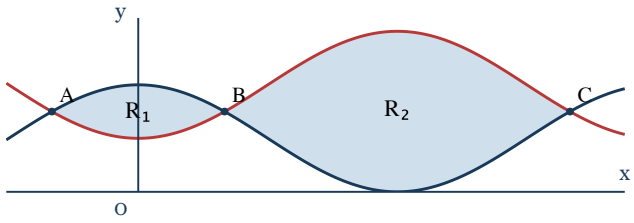
9. The curves are $y = 2\cos x + 2$ and $y = -2\cos x + 4$.

a. Find A, B, C.

b. Find R_1 in the form $a\sqrt{3} + b\pi/c$, with integer a, b, c.

c. Show $R_2 : R_1 = (3\sqrt{3} + 2\pi) : (3\sqrt{3} - \pi)$.

Blue: y = 2 cos x + 2; red: y = −2 cos x + 4



9a

Intersections satisfy

$2\cos x + 2 = -2\cos x + 4 \Rightarrow 4\cos x = 2$.

$\cos x = \frac{1}{2} \Rightarrow x = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}$.

At each, $y = 3$:

$A(-\frac{\pi}{3}, 3),\ B(\frac{\pi}{3}, 3),\ C(\frac{5\pi}{3}, 3)$.

9b

$$\int_{-\pi/3}^{\pi/3} (2\cos x + 2)\,dx = [2\sin x + 2x]_{-\pi/3}^{\pi/3}$$

$$= \left(\sqrt{3} + \frac{2\pi}{3}\right) - \left(-\sqrt{3} - \frac{2\pi}{3}\right) = 2\sqrt{3} + \frac{4\pi}{3}.$$

$$\int_{-\pi/3}^{\pi/3} (-2\cos x + 4)\,dx = [-2\sin x + 4x]_{-\pi/3}^{\pi/3}$$

$$= \left(-\sqrt{3} + \frac{4\pi}{3}\right) - \left(\sqrt{3} - \frac{4\pi}{3}\right) = -2\sqrt{3} + \frac{8\pi}{3}.$$

$$R_1 = 2\sqrt{3} + \frac{4\pi}{3} - \left(-2\sqrt{3} + \frac{8\pi}{3}\right) = 4\sqrt{3} - \frac{4\pi}{3}.$$

$a = 4, \quad b = -4, \quad c = 3$.

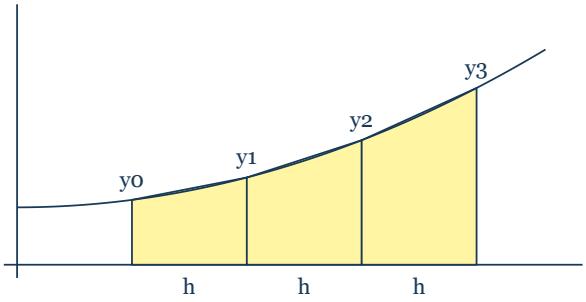
9c

$$R_2 = \int_{\pi/3}^{5\pi/3} (2 - 4\cos x)\,dx = 4\sqrt{3} + \frac{8\pi}{3}.$$

$R_2 : R_1 = (3\sqrt{3} + 2\pi) : (3\sqrt{3} - \pi)$.

The Trapezium Rule

Solution 69



Exact integration can be difficult. Approximate the area under the graph by dividing it into trapeziums of equal width h.

$$\begin{aligned}\text{Trapezium area} &= \frac{h}{2}(y_0 + y_1) \\ &\quad + \frac{h}{2}(y_1 + y_2) \\ &\quad + \frac{h}{2}(y_2 + y_3) \\ &= \frac{h}{2}[y_0 + 2(y_1 + y_2) + y_3].\end{aligned}$$

Trapezium Rule — Formula and Example

Solution 70

In general

$$\int_a^b y\,dx \approx \frac{h}{2}[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n].$$

The integral represents signed area (area when $y \geq 0$); $\approx$ means approximately equal; $h = (b-a)/n$ is the width of each trapezium.

Approximate the region bounded by $x = 1$, $x = 3$, the x-axis and $y = x^2$, using four strips.

The x-values:

x values

x	1	1.5	2	2.5	3

Dividing an interval of length 2 into four strips gives width 0.5.

The corresponding y-values:

y values

y	1	2.25	4	6.25	9

Width:

Width

$h = 0.5.$

Area:

Area

$$\text{Area} \approx \frac{0.5}{2}[1 + 2(2.25 + 4 + 6.25) + 9] = 8.75.$$

Trapezium Rule — Examination Example

Solution 71

Edexcel C2 May 2013 (R), Q2

$y = x/\sqrt{1+x}$.

a. Complete the table at $x = 1.3$, to 4 decimal places.

x	1	1.1	1.2	1.3	1.4	1.5
y	0.7071	0.7591	0.8090		0.9037	0.9487

Calculator tip: on the source Casio model, choose Mode → 3 (Table), use Alpha to enter X, then press =. Menu names vary by model.

a

At $x = 1.3$, $y = 0.8572$.

b. Use the trapezium rule with all table values to estimate $\int_1^{1.5} \frac{x}{\sqrt{1+x}}\,dx$, to 3 decimal places. Show every stage.

b

$h = 0.1$.

$$\text{Area} \approx \frac{0.1}{2}[0.7071 + 2(0.7591 + 0.8090 + 0.8572 + 0.9037) + 0.9487]$$

$$= 0.41569 \approx 0.416.$$

Further Example — Comparing Trapezium Estimates

Solution 72

Given $I = \int_0^{\pi/3} \sec x\,dx$:

a. Use $\int \sec x\,dx = \ln|\sec x + \tan x| + C$ to find I exactly.

b. Estimate I with two trapeziums.

c. Estimate I with four trapeziums.

d. Find the percentage error of each estimate.

a

$$I = [\ln|\sec x + \tan x|]_0^{\pi/3} = \ln(2+\sqrt{3}).$$

b

x	0	$\pi/6$	$\pi/3$
y	1	1.155	2

$$I \approx \frac{1}{2}\frac{\pi}{6}[1 + 2(1.155) + 2]$$

$$\approx 1.39.$$

c

x	0	$\pi/12$	$\pi/6$	$\pi/4$	$\pi/3$
y	1	1.035	1.155	1.414	2

$$I \approx \frac{1}{2}\frac{\pi}{12}[1 + 2(1.035 + 1.155 + 1.414) + 2]$$

$$\approx 1.34.$$

d

Use $100|T - I|/I$.

Keeping full calculator precision gives $T_2 = 1.38999795\ldots$, $T_4 = 1.33627389\ldots$.

Percentage errors: approximately 5.55% and 1.47%.

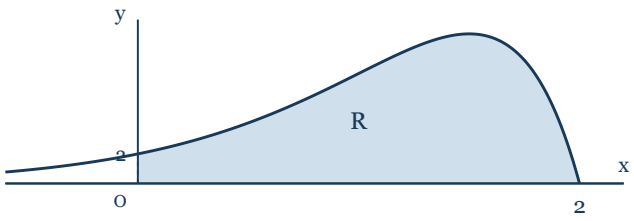
Using the rounded estimates 1.39 and 1.34 instead gives approximately 5.5% and 1.7%.

Test Your Understanding — Numerical and Exact Area

Solution 73

Edexcel C4 June 2014 (R), Q2

$y = (2 - x)e^{2x}$. Region R is bounded by the curve and the coordinate axes.



x	0	0.5	1	1.5	2
y	2	4.077	7.389	10.043	0

a. Use all table values in the trapezium rule to approximate the area, to 2 d.p.

b. Explain how to improve the approximation.

c. Use calculus to find the exact area in simplest form, showing each step.

a

$$\text{Area} \approx \frac{1}{2}(0.5)[2 + 2(4.077 + 7.389 + 10.043) + 0]$$

$$= \frac{1}{4}(45.018) = 11.2545 \approx 11.25.$$

b

Increase the number of strips: use more trapeziums, smaller h, and more equally spaced x- and y-values.

c

$$u = 2 - x, \quad \frac{du}{dx} = -1; \qquad \frac{dv}{dx} = e^{2x}, \quad v = \frac{1}{2}e^{2x}.$$

$$\int (2 - x)e^{2x}\,dx = \frac{1}{2}(2 - x)e^{2x} - \int -\frac{1}{2}e^{2x}\,dx$$

$$= \frac{1}{2}(2 - x)e^{2x} + \frac{1}{4}e^{2x} + C.$$

$$\text{Area} = \left[\frac{1}{2}(2 - x)e^{2x} + \frac{1}{4}e^{2x}\right]_0^2$$

$$= \left(0 + \frac{1}{4}e^4\right) - \left(\frac{1}{2}(2)e^0 + \frac{1}{4}e^0\right) = \frac{1}{4}e^4 - \frac{5}{4}.$$

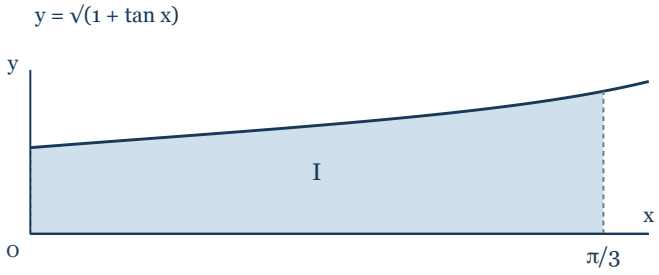
Exercise 11I — Pages 319–322

Solution 74

1. $y = \sqrt{1 + \tan x}$, $0 \le x \le \pi/3$.

a. Complete the missing table entries.

x	0	$\pi/12$	$\pi/6$	$\pi/4$	$\pi/3$
y	1		1.2559		1.6529



1a

At $x = \pi/12$, $y = 1.1260$.

At $x = \pi/4$, $y = 1.4142$.

Given $I = \int_0^{\pi/3} \sqrt{1 + \tan x}\,dx$, use the trapezium rule:

b(i). With $x = 0, \pi/6, \pi/3$, estimate I to 4 s.f.

b(ii). With all five table values, estimate I to 4 s.f.

1b

$$\text{Area} \approx \frac{h}{2}[y_0 + 2(y_1 + \cdots) + y_n].$$

$$T_2 = \frac{1}{2}\frac{\pi}{6}(1 + 1.6529 + 2 \times 1.2559)$$

$$\approx 1.352.$$

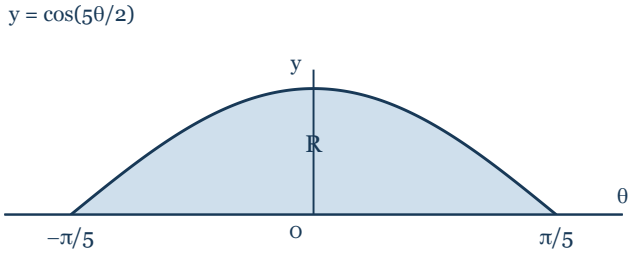
$$T_4 = \frac{\pi}{24}\big[1 + 1.6529 + 2(1.1260 + 1.2559 + 1.4142)\big]$$

$$\approx 1.341.$$

Exercise 11I — Pages 319–322 Solution 75

2. Region R lies between $y = \cos(5\theta/2)$ and the horizontal axis, for $-\pi/5 \le \theta \le \pi/5$.

θ	$-\pi/5$	$-\pi/10$	0	$\pi/10$	$\pi/5$
y	0		1		0



a. Complete the table to 4 d.p.

b. Use all table values to estimate the area with the trapezium rule, to 3 d.p.

c. State whether this is an underestimate or overestimate, with a reason.

d. Integrate to find the exact area.

e. Calculate the percentage error in part b.

2a

The two missing values are both 0.7071.

2b

$$R \approx \frac{1}{2}\frac{\pi}{10}[0 + 0 + 2(0.7071 + 1 + 0.7071)]$$
$$\approx 0.758.$$

2c

The graph is concave down, so the straight trapezium tops lie below the curve: this is an underestimate.

2d

$$\int_{-\pi/5}^{\pi/5} \cos\frac{5\theta}{2}\,d\theta = \left[\frac{2}{5}\sin\frac{5\theta}{2}\right]_{-\pi/5}^{\pi/5}$$
$$= \frac{4}{5} = 0.8.$$

2e

Using the answer from part b:

$$\text{Percentage error} = \frac{0.8 - 0.758}{0.8} \times 100\% = 5.25\%.$$

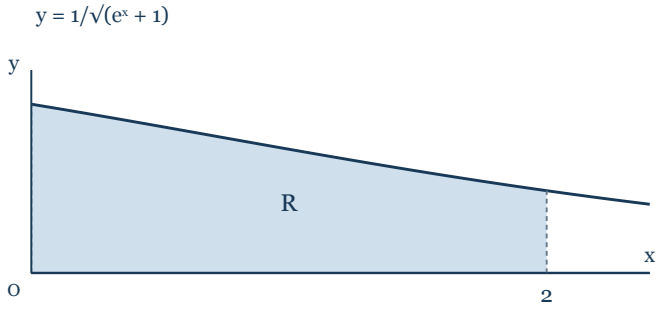
Exercise 11I — Pages 319–322 Solution 76

3. $y = 1/\sqrt{e^x + 1}$. Region R is bounded by the curve, both coordinate axes and $x = 2$.

a. Complete the table to 3 d.p.

x	0	0.5	1	1.5	2
y	0.707	0.614	0.519		0.345

b. Use all table values in the trapezium rule to estimate the area of R, to 2 d.p.



3a

At $x = 1.5$, $y = 0.427$.

3b

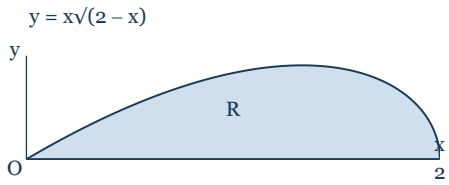
$$\text{Area} \approx \frac{h}{2}[y_0 + 2(y_1 + \cdots) + y_n]$$
$$= \frac{1}{4}[0.707 + 0.345 + 2(0.614 + 0.519 + 0.427)]$$
$$= 1.043 \approx 1.04.$$

Exercise 11I — Pages 319–322 — Solution 77

5. $y = x\sqrt{2-x}$, $0 \le x \le 2$.

a. Complete the table at $x = 1.5$.

x	0	0.5	1	1.5	2
y	0	0.6124	1		0



Given $I = \int_0^2 x\sqrt{2-x}\,dx$:

b. Use four trapeziums to estimate I, to 4 s.f.

c. Use a substitution or otherwise to find I exactly in the form $2^q p$, for rational p, q.

d. Calculate the percentage error of part b.

5a

At $x = 1.5$, $y = 1.0607$.

5b

$$I \approx \frac{1}{2}(0.5)[0 + 0 + 2(0.6124 + 1 + 1.0607)]$$
$$= 1.33655 \approx 1.337.$$

5c

Let $u = 2 - x$, so $du/dx = -1$.

$$I = \int_2^0 -(2-u)u^{1/2}\,du = \int_2^0 (u^{3/2} - 2u^{1/2})\,du$$
$$= \left[\frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2}\right]_2^0$$
$$= 0 - \left(\frac{2}{5}\sqrt{32} - \frac{4}{3}\sqrt{8}\right)$$
$$= \frac{8}{3}\sqrt{2} - \frac{8}{5}\sqrt{2} = \frac{16}{15}\sqrt{2} = \frac{2^{9/2}}{15}.$$
$$p = \frac{16}{15}, \quad q = \frac{1}{2}.$$

5d

$$I = \frac{16}{15}\sqrt{2} \approx 1.509.$$
$$\text{Percentage error} = \frac{I - 1.337}{I} \times 100\% \approx 11.4\%.$$

Exercise 11I — Pages 319–322 — Solution 78

7. $I = \int_0^3 e^{\sqrt{2x+1}}\,dx$.

a. Complete the table for $y = e^{\sqrt{2x+1}}$.

x	0	0.5	1	1.5	2	2.5	3
y	2.7183				9.3565	11.5824	14.0940

b. Use all table values in the trapezium rule to estimate I, to 4 s.f.

c. Use $t = \sqrt{2x+1}$ to express I as $\int_a^b kte^t\,dt$, finding a, b, k.

d. Integrate by parts and find I, to 4 s.f.

7a The missing values are 4.1133, 5.6522, 7.3891.

7b

$$I \approx \frac{1}{4}\big[2.7183 + 14.0940 + 2(4.1133 + 5.6522 + 7.3891 + 9.3565 + 11.5824)\big]$$
$$\approx 23.25.$$

7c

$$t = \sqrt{2x+1} \Rightarrow \frac{dt}{dx} = \frac{1}{\sqrt{2x+1}} = \frac{1}{t}.$$
$$I = \int_1^{\sqrt{7}} te^t\,dt.$$
$$a = 1, \quad b = \sqrt{7}, \quad k = 1.$$

7d

$$u = t \Rightarrow \frac{du}{dt} = 1; \qquad \frac{dv}{dt} = e^t \Rightarrow v = e^t.$$
$$I = [te^t]_1^{\sqrt{7}} - \int_1^{\sqrt{7}} e^t\,dt$$
$$= \sqrt{7}e^{\sqrt{7}} - e - e^{\sqrt{7}} + e$$
$$= (\sqrt{7} - 1)e^{\sqrt{7}} \approx 23.20.$$

Integration with Parametric Equations

Solution 79

Suppose $x = t^2$ and $y = t + 1$. To find area we need $\int y\,dx$, but y is expressed in terms of t.

Rewrite the integral:

$$\int y\,dx = \int y\frac{dx}{dt}\,dt.$$

Memory tip: $dx = (dx/dt)\,dt$, the substitution rule derived from the chain rule. Informally the dt factors appear to cancel.

Find the area bounded by $x = t^2$, $y = t + 1$, $t > 0$, the x-axis and $x = 0$, $x = 3$.

Step 1: find dx/dt.

Step 2: change the limits to t-values.

Step 3: integrate.

Derivative

$$\frac{dx}{dt} = 2t.$$

Limits

Since we are integrating with respect to t, change the limits to t-values.

When $x = 3$, $t = \sqrt{3}$.

The boundary $x = 0$ corresponds to $t = 0$.

Integrate

$$A = \int_0^{\sqrt{3}} y\frac{dx}{dt}\,dt = \int_0^{\sqrt{3}} (t+1)2t\,dt$$

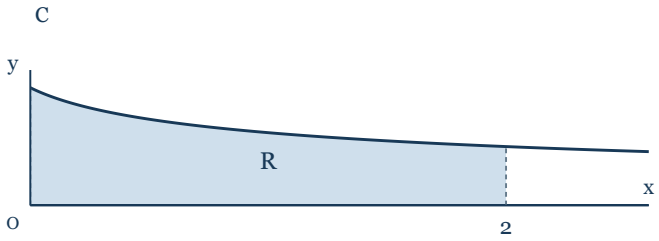
$$= \int_0^{\sqrt{3}} (2t^2 + 2t)\,dt$$

$$= \left[\frac{2}{3}t^3 + t^2\right]_0^{\sqrt{3}} = 2\sqrt{3} + 3.$$

Further Parametric Area Example

Solution 80

Curve C has parametric equations $x = t(1 + t)$, $y = 1/(1 + t)$, $t \geq 0$. Find the exact area R, bounded by C, the x-axis and $x = 0$, $x = 2$.



Derivative

$$\frac{dx}{dt} = 1 + 2t.$$

Lower limit

When $x = 0$:

$0 = t(1 + t) \Rightarrow t = 0$ or -1.

Use $t = 0$, since $t \geq 0$.

Upper limit

When $x = 2$:

$2 = t(1 + t) \Rightarrow t^2 + t - 2 = 0 \Rightarrow t = -2$ or 1.

Use $t = 1$.

Integral

$$R = \int_0^1 y\frac{dx}{dt}\,dt = \int_0^1 \frac{1}{1+t}(1 + 2t)\,dt$$

$$= \int_0^1 \frac{1 + 2t}{1 + t}\,dt = \int_0^1 \left(2 - \frac{1}{1+t}\right)dt$$

$$= [2t - \ln|1 + t|]_0^1 = 2 - \ln 2.$$

Recall: divide an improper fraction before integrating.

Test Your Understanding — Parametric Area

Solution 81

Edexcel C4 January 2013, Q5

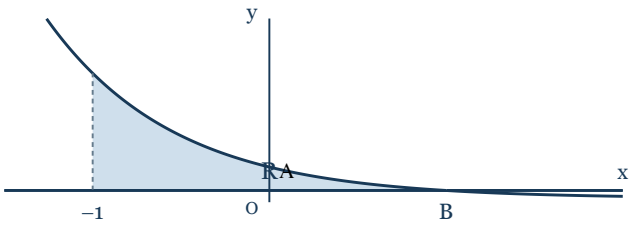
Curve C: $x = 1 - \frac{1}{2}t$, $y = 2^t - 1$. It crosses the y-axis at A and the x-axis at B.

a. Show that $A = (0, 3)$.

b. Find the x-coordinate of B.

d. Find the exact area R, bounded by C, $x = -1$ and the x-axis.

C; A is the y-intercept



Tip: $\frac{d}{dx}a^x = a^x \ln a$, and $\int a^x\,dx = a^x / \ln a + C$ for $a > 0$, $a \neq 1$.

a

At $x = 0$: $0 = 1 - \frac{1}{2}t \Rightarrow t = 2$.

Then $y = 2^2 - 1 = 3$, so $A = (0, 3)$.

b

At $y = 0$: $2^t = 1 \Rightarrow t = 0$.

Hence $x = 1 - \frac{1}{2}(0) = 1$.

d

$$\text{Area}(R) = \int (2^t - 1)(-\tfrac{1}{2})\,dt.$$

Limits: $x = -1 \Rightarrow t = 4$; $x = 1 \Rightarrow t = 0$.

$$R = -\frac{1}{2}\left[\frac{2^t}{\ln 2} - t\right]_4^0$$

$$= -\frac{1}{2}\left[\frac{1}{\ln 2} - \left(\frac{16}{\ln 2} - 4\right)\right]$$

$$= \frac{15}{2\ln 2} - 2.$$

Parametric Area — Question 1 (source slide 82)

Solution 82

1. Curve C has parametric equations $x = t^3$, $y = t^2$, $t \geq 0$. Show that the exact area bounded by the curve, the x-axis and $x = 0$, $x = 4$ is $k\sqrt[3]{2}$, where k is a rational constant to be found.

1

$x = 0 \Rightarrow t = 0$, $\quad x = 4 \Rightarrow t = \sqrt[3]{4}$.

$$\frac{dx}{dt} = 3t^2$$

$$A = \int_0^{\sqrt[3]{4}} t^2(3t^2)\,dt = \left[\frac{3}{5}t^5\right]_0^{\sqrt[3]{4}}$$

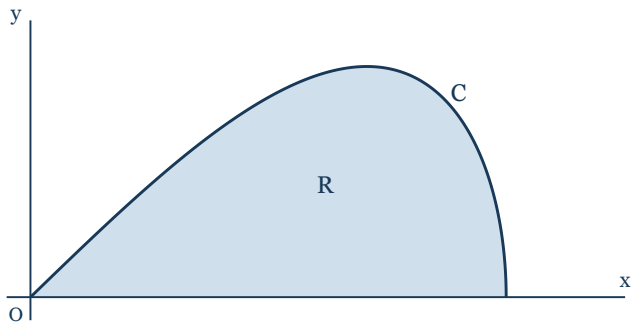
$$= \frac{3}{5}(\sqrt[3]{4})^5 = \frac{24}{5}\sqrt[3]{2}.$$

$$k = \frac{24}{5}.$$

Parametric Area — Question 2 (source slide 82)

Solution 83

2. Curve C has parametric equations $x = \sin t,\ y = \sin 2t,\ 0 \le t \le \pi/2$. The finite region R is bounded by the curve and the x-axis. Find its exact area. (6 marks)



2

$$\frac{dx}{dt} = \cos t$$

$$A = \int_0^{\pi/2} \sin 2t \cos t \, dt = \int_0^{\pi/2} 2 \sin t \cos^2 t \, dt$$

$$= \left[-\frac{2}{3} \cos^3 t \right]_0^{\pi/2} = \frac{2}{3}.$$

Parametric Area — Question 3 (source slide 82)

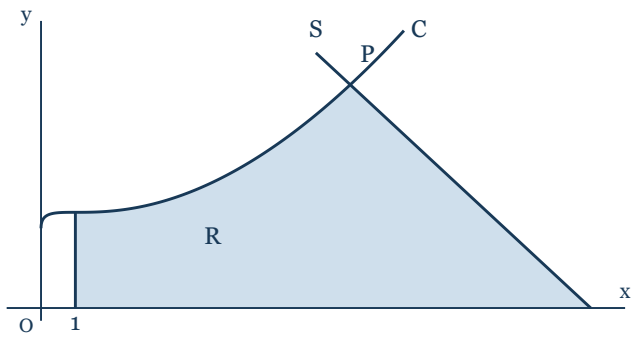
Solution 84

3. Curve C has parametric equations $x = (t+1)^2,\ y = \frac{1}{2}t^3 + 3,\ t \ge -1$. Point P has $t = 2$, and S is the normal at P.

(a) Find an equation of S. (5 marks)

Region R is bounded by C, S, the x-axis and $x = 1$.

(b) Use integration to find its area. (5 marks)



3a

$$P = (9, 7), \quad \frac{dy}{dx} = \frac{3t^2/2}{2(t+1)} = 1 \text{ at } t = 2.$$

Normal gradient is -1:

$$y - 7 = -(x - 9) \Rightarrow x + y = 16.$$

3b

$x = 1 \Rightarrow t = 0$, S meets the axis at $x = 16$.

$$A = \int_0^2 (\tfrac{1}{2}t^3 + 3)\, 2(t+1)\, dt + \tfrac{1}{2}(16 - 9)7$$

$$= \left[\frac{t^5}{5} + \frac{t^4}{4} + 3t^2 + 6t \right]_0^2 + \frac{49}{2}$$

$$= \frac{172}{5} + \frac{49}{2} = \frac{589}{10} = 58.9.$$

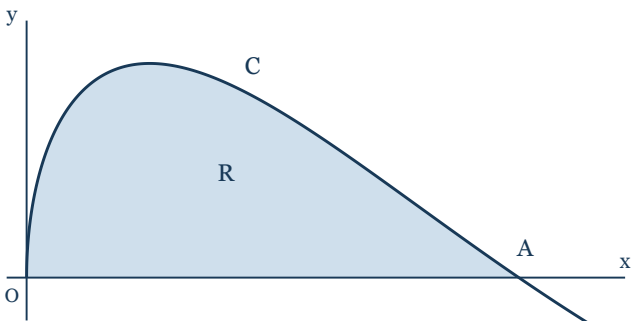
Parametric Area — Question 4 (source slide 82)

Solution 85

4. Curve C has parametric equations $x = 3t^2,\ y = \sin 2t,\ t \geq 0$.

(a) State t at point A, the first positive crossing of the x-axis. (1 mark)

(b) Find, in terms of π, the exact shaded area bounded by C and the x-axis. (6 marks)



4a $\sin 2t = 0 \Rightarrow t = \dfrac{\pi}{2}$ at A.

4b

$$A = \int_0^{\pi/2} \sin 2t\,(6t)\,dt$$

$$= \left[-3t\cos 2t + \frac{3}{2}\sin 2t\right]_0^{\pi/2}$$

$$= \frac{3\pi}{2}.$$

Parametric Area — Question 5 (source slide 82)

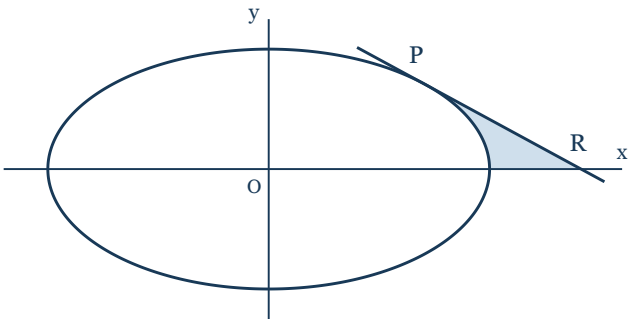
Solution 86

5. Curve C has parametric equations $x = 5\cos\theta,\ y = 4\sin\theta,\ 0 \leq \theta < 2\pi$. At P, $\theta = \pi/4$.

(a) Find the gradient at P. (3 marks)

(b) Find the tangent equation at P. (3 marks)

(c) Find the exact shaded area bounded by tangent PR, the curve and the x-axis. (6 marks)



5a $\dfrac{dy}{dx} = \dfrac{4\cos\theta}{-5\sin\theta} = -\dfrac{4}{5}$ at P.

5b

$P = (5/\sqrt{2}, 2\sqrt{2})$

$y - 2\sqrt{2} = -\dfrac{4}{5}(x - 5/\sqrt{2})$

$y = 4\sqrt{2} - \dfrac{4}{5}x, \quad R = (5\sqrt{2}, 0).$

5c

Triangle under the tangent:

$\frac{1}{2}(5\sqrt{2} - 5/\sqrt{2})(2\sqrt{2}) = 5.$

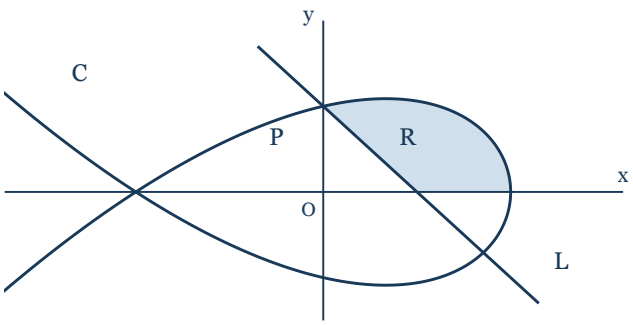
$$A = 5 - \int_0^{\pi/4} 20\sin^2\theta\,d\theta$$

$$= 5 - [10\theta - 5\sin 2\theta]_0^{\pi/4} = 10 - \frac{5\pi}{2}.$$

Parametric Area — Question 6 (source slide 82)

Solution 87

6. Curve C has parametric equations $x = 1 - t^2,\ y = 2t - t^3,\ t \in \mathbb{R}$. Line L is the normal at P, where C intersects the positive y-axis. Find the exact area of R, bounded by C, L and the x-axis, as shown. (7 marks)



6

$x = 0 \Rightarrow t = \pm 1; \quad P = (0, 1)$ uses $t = 1$.

$\dfrac{dy}{dx} = \dfrac{2 - 3t^2}{-2t} = \dfrac{1}{2}$ at P.

Normal: $y = 1 - 2x$, meeting the axis at $x = \frac{1}{2}$.

$$A = \int_1^0 (2t - t^3)(-2t)\,dt - \frac{1}{2} \cdot \frac{1}{2} \cdot 1$$

$$= \left[\frac{4}{3}t^3 - \frac{2}{5}t^5\right]_0^1 - \frac{1}{4}$$

$$= \frac{14}{15} - \frac{1}{4} = \frac{41}{60}.$$

Parametric Area — Question 7 (source slide 82)

Solution 88

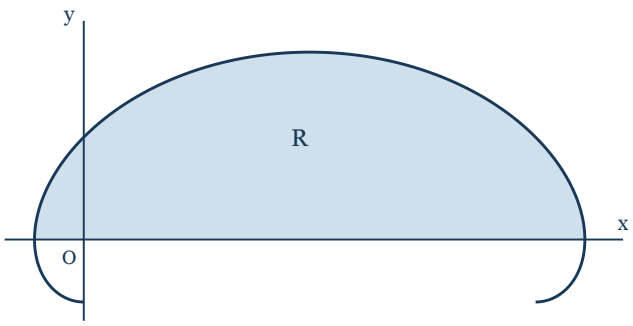
7. Curve C has parametric equations $x = t - 2\sin t,\ y = 1 - 2\cos t,\ 0 \le t \le 2\pi$.

(a) Show it crosses the x-axis at $t = \pi/3, 5\pi/3$. (3 marks)

Region R is enclosed by C and the x-axis.

(b) Show its area is $\int_{\pi/3}^{5\pi/3} (1 - 2\cos t)^2\,dt$. (3 marks)

(c) Evaluate this integral exactly. (4 marks)



7a

$1 - 2\cos t = 0 \Rightarrow \cos t = \dfrac{1}{2}$

$t = \dfrac{\pi}{3}$ or $\dfrac{5\pi}{3}$.

7b

$\dfrac{dx}{dt} = 1 - 2\cos t$

$$A = \int_{\pi/3}^{5\pi/3} y\frac{dx}{dt}\,dt = \int_{\pi/3}^{5\pi/3} (1 - 2\cos t)^2\,dt.$$

7c

$(1 - 2\cos t)^2 = 3 - 4\cos t + 2\cos 2t$

$A = [3t - 4\sin t + \sin 2t]_{\pi/3}^{5\pi/3}$

$= 4\pi + 3\sqrt{3}.$

Differential Equations — Separating Variables

Solution 89

Differential equations contain variables and derivatives, for example x, y and dy/dx. Solving means expressing the relationship between the variables without a derivative remaining.

Find the general solution of $dy/dx = xy + y$.

Step 1

Factorise if needed, then separate: put the y-terms with dy/dx, and the x-terms on the other side.

Separate

$$\frac{dy}{dx} = y(x+1), \qquad \frac{1}{y}\frac{dy}{dx} = x+1 \quad (y \neq 0).$$

Step 2

Integrate. The chain rule lets $(dy/dx)\,dx$ become dy.

Integrate

$$\int \frac{1}{y}\frac{dy}{dx}\,dx = \int (x+1)\,dx$$

$$\int \frac{1}{y}\,dy = \int (x+1)\,dx$$

$$\ln|y| = \frac{1}{2}x^2 + x + C.$$

The intermediate integral line can be skipped once the substitution is clear.

Step 3

Make y the subject if asked.

Subject

$$|y| = e^{x^2/2+x+C} = e^{x^2/2+x}e^C.$$

$$y = Ae^{x^2/2+x}.$$

For nonzero solutions $A = \pm e^C$. The solution $y = 0$ is also allowed, so the general constant A may be any real number.

Tip: combine $e^{f(x)+C}$ as $e^C e^{f(x)}$. The factor e^C is positive; a sign can be absorbed into a new constant when undoing $\ln|y|$.

Another Separable Differential Equation

Solution 90

Find the general solution of $(1+x^2)\frac{dy}{dx} = x\tan y$.

Step 1

Separate the variables.

Separate

For $\tan y \neq 0$:

$$\frac{1}{\tan y}\frac{dy}{dx} = \frac{x}{1+x^2}.$$

Step 2

Integrate.

Integrate

$$\int \cot y\,dy = \int \frac{x}{1+x^2}\,dx$$

$$\ln|\sin y| = \frac{1}{2}\ln(1+x^2) + C.$$

Step 2b

Combine the constant with the logarithms.

Constant

Write $C = \ln k$, with $k > 0$:

$$\ln|\sin y| = \ln[k\sqrt{1+x^2}].$$

$$\sin y = K\sqrt{1+x^2}, \quad K = \pm k.$$

Step 3

Make y the subject on a chosen branch.

Subject

$$y = n\pi + (-1)^n \arcsin(K\sqrt{1+x^2}), \quad n \in \mathbb{Z}.$$

The principal branch is $y = \arcsin(K\sqrt{1+x^2})$. Keep the integer n fixed on each connected interval. Use intervals where $|K|\sqrt{1+x^2} < 1$, since $\tan y$ must be defined.

Taking $K = 0$ also includes the constant solutions $y = n\pi$, lost when dividing by $\tan y$.

Differential Equations with a Given Condition

Solution 91

Solve $\frac{dy}{dx} = -\frac{3(y-2)}{(2x+1)(x+2)}$, given $y = 4$ when $x = 1$.
Leave the answer as $y = f(x)$.

Working

Tip: apply the condition as soon as it makes the constant easy to determine.

$$\int \frac{1}{y-2}\,dy = \int \frac{-3}{(2x+1)(x+2)}\,dx.$$

Split the right side into partial fractions:

$$\frac{-3}{(2x+1)(x+2)} = \frac{A}{2x+1} + \frac{B}{x+2}.$$

$$A = -2, \qquad B = 1.$$

$$\int \frac{1}{y-2}\,dy = \int \left(\frac{1}{x+2} - \frac{2}{2x+1}\right) dx$$

$$\ln|y-2| = \ln|x+2| - \ln|2x+1| + C$$

$$= \ln\left|\frac{k(x+2)}{2x+1}\right| \quad (k>0).$$

Absorb any sign into the constant:

$$y - 2 = \frac{K(x+2)}{2x+1}.$$

At $(1,4)$: $4-2 = K(1+2)/(2+1) \Rightarrow K = 2$.

$$y = 2 + \frac{2(x+2)}{2x+1} = 3 + \frac{3}{2x+1}, \qquad x > -\frac{1}{2}.$$

Test Your Understanding — Differential Equations

Solution 92

Edexcel C4 January 2012, Q4

Given $y = 2$ at $x = \pi/4$, solve

$$\frac{dy}{dx} = \frac{3}{y\cos^2 x}.$$

Working

$$\int y\,dy = \int \frac{3}{\cos^2 x}\,dx = \int 3\sec^2 x\,dx.$$

$$\frac{1}{2}y^2 = 3\tan x + C.$$

At $y = 2$, $x = \pi/4$:

$$\frac{1}{2}(2)^2 = 3\tan\frac{\pi}{4} + C \quad \Rightarrow \quad C = -1.$$

$$\frac{1}{2}y^2 = 3\tan x - 1.$$

The given positive value selects

$$y = \sqrt{6\tan x - 2}, \qquad \arctan(1/3) < x < \pi/2.$$

Key Points on Differential Equations

Solution 93

1. Separate variables: put the y-terms on the left by dividing, factorising first if necessary. Check whether dividing loses constant solutions.
2. When logarithms occur, it can help to write the integration constant as $\ln k$, with $k > 0$.
3. Combine logarithms using their laws. For example:

$$2\ln|x+1| - \ln|x| \quad \longrightarrow$$

$$\ln\left|\frac{(x+1)^2}{x}\right|.$$

4. Substitute the given conditions to determine the constant; it is often easier to do this early.
5. Examination questions often combine partial fractions with differential equations.

Exercise 11J — Pages 324–326

Solution 94

4. Given $x = 0$ when $y = 0$, find a particular solution of $(2y + 2yx)\frac{dy}{dx} = 1 - y^2$, in the form $y = g(x)$.

4

Formal separation gives

$$2y(1+x)\frac{dy}{dx} = 1 - y^2$$

$$\int \frac{2y}{1-y^2}\,dy = \int \frac{1}{1+x}\,dx$$

$$\ln k - \ln|1-y^2| = \ln|1+x|$$

$$\ln\left|\frac{k}{1-y^2}\right| = \ln|1+x|.$$

Using the proposed condition formally gives $k = 1$:

$$\frac{1}{1-y^2} = 1 + x, \quad 1 - y^2 = \frac{1}{x+1}, \quad y^2 = \frac{x}{x+1}.$$

However, at $(0, 0)$ the original equation is $0 = 1$. There is no differentiable solution through this point. The branches $y = \pm\sqrt{x/(x+1)}$, $x > 0$, approach it but have no finite derivative there.

5. Find the general solution of $e^{x+y}\frac{dy}{dx} = 2x + xe^y$, in the form $\ln|g(y)| = f(x)$.

5

$$e^x e^y \frac{dy}{dx} = x(2 + e^y).$$

$$\frac{e^y}{2+e^y}\frac{dy}{dx} = xe^{-x}.$$

$$\int \frac{e^y}{2+e^y}\,dy = \int xe^{-x}\,dx$$

$$\ln|2+e^y| = \int xe^{-x}\,dx.$$

$$u = x, \quad \frac{du}{dx} = 1; \qquad \frac{dv}{dx} = e^{-x}, \quad v = -e^{-x}.$$

$$\ln|2+e^y| = -xe^{-x} + \int e^{-x}\,dx$$

$$= -xe^{-x} - e^{-x} + C.$$

6. Solve $(1-x^2)\frac{dy}{dx} = xy + y$, given $y = 6$ at $x = 0.5$. Give $y = f(x)$.

7. Solve $(1+x^2)\frac{dy}{dx} = x - xy^2$, given $y = 2$ at $x = 0$. Give $y = f(x)$.

6

$$(1-x^2)\frac{dy}{dx} = y(x+1).$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{x+1}{1-x^2} = \frac{x+1}{(1-x)(1+x)} = \frac{1}{1-x}.$$

$$\int \frac{1}{y}\,dy = \int \frac{1}{1-x}\,dx.$$

$$\ln|y| = \ln k - \ln|1-x| = \ln\left|\frac{k}{1-x}\right|.$$

Absorbing the sign into K: $y = K/(1-x)$.

At $(0.5, 6)$, $6 = 2K \Rightarrow K = 3$.

$$y = \frac{3}{1-x}, \qquad x < 1.$$

The formula also satisfies the original equation at $x = -1$, where the intermediate cancellation was not valid.

7

$$(1+x^2)\frac{dy}{dx} = x(1-y^2).$$

$$\int \frac{1}{1-y^2}\,dy = \int \frac{x}{1+x^2}\,dx$$

$$\frac{1}{2}\int\left(\frac{1}{1+y} + \frac{1}{1-y}\right)dy = \int \frac{x}{1+x^2}\,dx$$

$$\frac{1}{2}(\ln|1+y| - \ln|1-y|) = \frac{1}{2}\ln(1+x^2) + C.$$

At $(0, 2)$, $C = \frac{1}{2}\ln 3$.

$$\ln|1+y| - \ln|1-y| = \ln[3(1+x^2)].$$

$$\frac{1+y}{y-1} = 3 + 3x^2 \Rightarrow 1 + y = 3y + 3x^2y - 3 - 3x^2.$$

$$1 + 3 + 3x^2 = 3y + 3x^2y - y$$

$$3(x^2+1) + 1 = y[3(x^2+1) - 1].$$

$$y = \frac{3(1+x^2)+1}{3(1+x^2)-1}.$$

8. Solve $dy/dx = xe^{-y}$, given $y = \ln 2$ at $x = 4$. Give $y = f(x)$.

9. Solve $dy/dx = \cos^2 y + \cos 2x \cos^2 y$, given $y = \pi/4$ at $x = \pi/4$. Give $\tan y = f(x)$.

8

$$\int e^y\,dy = \int x\,dx \quad \Rightarrow \quad e^y = \frac{x^2}{2} + C.$$

At $(4, \ln 2)$: $2 = 8 + C \Rightarrow C = -6$.

$$e^y = \frac{x^2 - 12}{2}.$$

$$y = \ln\left(\frac{x^2-12}{2}\right), \qquad x > \sqrt{12}.$$

The logarithm argument must be positive; absolute values must not introduce a false extra branch.

9

$$\frac{dy}{dx} = \cos^2 y(1 + \cos 2x).$$

$$\int \sec^2 y\,dy = \int (1 + \cos 2x)\,dx.$$

$$\tan y = x + \frac{1}{2}\sin 2x + C.$$

At $(\pi/4, \pi/4)$:

$$1 = \frac{\pi}{4} + \frac{1}{2}\sin\frac{\pi}{2} + C$$

$$C = \frac{1}{2} - \frac{\pi}{4} = \frac{2-\pi}{4}.$$

$$\tan y = x + \frac{1}{2}\sin 2x + \frac{2-\pi}{4}.$$

Exercise 11J — Pages 324–326 Solution 97

10. Given $y = 1$ at $x = \pi/2$, solve $dy/dx = xy\sin x$.

10

$$\int \frac{1}{y}\,dy = \int x\sin x\,dx.$$

$$\ln|y| = -x\cos x + \int \cos x\,dx$$

$$= -x\cos x + \sin x + C.$$

At $(\pi/2, 1)$, $C = -1$.

$$\ln|y| = -x\cos x + \sin x - 1.$$

$$y = e^{-x\cos x + \sin x - 1}.$$

11a. Find $\int \frac{3x+4}{x}\,dx$, $x > 0$.

11a

$$\int \left(3 + \frac{4}{x}\right) dx = 3x + 4\ln x + C.$$

11b. Given $y = 16$ at $x = 1$, solve

$$\frac{dy}{dx} = \frac{3x\sqrt{y} + 4\sqrt{y}}{x},$$

giving $y = f(x)$.

11b

$$\frac{dy}{dx} = \sqrt{y}\,\frac{3x+4}{x}.$$

$$\int \frac{1}{\sqrt{y}}\,dy = 3x + 4\ln x + C.$$

$$2\sqrt{y} = 3x + 4\ln x + C.$$

At $(1, 16)$: $8 = 3 + C \Rightarrow C = 5$.

$$\sqrt{y} = \frac{3}{2}x + 2\ln x + \frac{5}{2}.$$

$$y = \left(\frac{3}{2}x + 2\ln x + \frac{5}{2}\right)^2.$$

This is the positive branch. Use the interval containing $x = 1$ on which $x > 0$ and the expression before squaring is positive.

Exercise 11J — Pages 324–326 Solution 98

12a. Express $\frac{8x-18}{(3x-8)(x-2)}$ in partial fractions.

b. For $x \geq 3$, find the general solution of

$$(x-2)(3x-8)\frac{dy}{dx} = (8x-18)y.$$

c. Hence find the particular solution with $y = 8$ at $x = 3$, as $y = f(x)$.

12a

$$\frac{8x-18}{(3x-8)(x-2)} = \frac{A}{3x-8} + \frac{B}{x-2}.$$

$$8x - 18 = A(x-2) + B(3x-8).$$

$$x = 2: \quad -2 = -2B \Rightarrow B = 1.$$

$$x = \tfrac{8}{3}: \quad \tfrac{64}{3} - 18 = \tfrac{2}{3}A \Rightarrow A = 5.$$

$$\frac{8x-18}{(3x-8)(x-2)} = \frac{5}{3x-8} + \frac{1}{x-2}.$$

12b

$$\int \frac{1}{y}\,dy = \int \frac{8x-18}{(3x-8)(x-2)}\,dx$$

$$= \int \left(\frac{5}{3x-8} + \frac{1}{x-2}\right) dx.$$

$$\ln|y| = \frac{5}{3}\ln|3x-8| + \ln|x-2| + C.$$

$$y = K(x-2)(3x-8)^{5/3}, \quad K \in \mathbb{R}.$$

This includes $y = 0$.

12c

At $(3, 8)$, $C = \ln 8$, and the positive branch is selected.

$$\ln y = \ln[(3x-8)^{5/3}] + \ln[8(x-2)]$$

$$= \ln[8(x-2)(3x-8)^{5/3}].$$

$$y = 8(x-2)(3x-8)^{5/3}.$$

Exercise 11J — Pages 324–326 Solution 99

14a. Find the general solution of $dy/dx = -1/(x+2)^2$.

b. On the same axes, sketch three particular solutions.

c. Write down the solution through $(8, 3.1)$.

14a

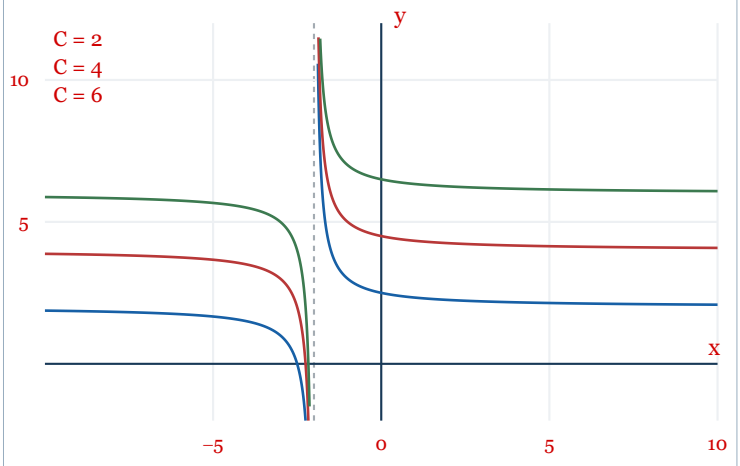
$$y = -\int \frac{1}{(x+2)^2}\,dx = \frac{1}{x+2} + C, \quad x \neq -2.$$

14c

$$3.1 = \frac{1}{10} + C \Rightarrow C = 3.$$

$$y = \frac{1}{x+2} + 3.$$

14b



14b explanation

Three examples use $C = 2, 4, 6$. Any real constant gives a vertical translate; $x = -2$ is a vertical asymptote.

Exercise 11J — Pages 324–326 Solution 100

15a. Show that the general solution of $dy/dx = -x/y$ can be written $x^2 + y^2 = C$.

b. Sketch three particular solution curves on the same axes.

c. Write down the solution through $(0, 7)$.

15a

$$\int y\,dy = -\int x\,dx$$

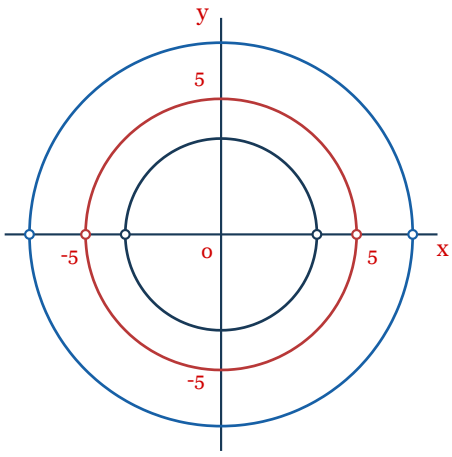
$$\frac{y^2}{2} = -\frac{x^2}{2} + K \quad \Rightarrow \quad x^2 + y^2 = C.$$

15c

$$x^2 + y^2 = 49.$$

The branch through $(0, 7)$ is $y = \sqrt{49 - x^2}$, $-7 < x < 7$.

15b



15b explanation

The implicit curves are circles centred at $(0, 0)$ with radius $\sqrt{C}$, for $C > 0$; examples shown: $C = 10, 20, 40$.

Each solution $y(x)$ follows an upper or lower arc. The equation is undefined at $y = 0$, shown by open points.

Exercise 11J — Extension

Solution 101

Extension — STEP I 2013, Q7

i. Use $y = ux$, where u is a function of x, to show that the solution of

$$\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}, \qquad x > 0,\ y > 0,$$

with $y = 2$ when $x = 1$, is

$$y = x\sqrt{4 + 2\ln x}, \qquad x > e^{-2}.$$

ii. Use a substitution to solve

$$\frac{dy}{dx} = \frac{x}{y} + \frac{2y}{x}, \qquad x > 0,\ y > 0,$$

with $y = 2$ at $x = 1$.

iii. Solve

$$\frac{dy}{dx} = \frac{x^2}{y} + \frac{2y}{x}, \qquad x > 0,\ y > 0,$$

with $y = 2$ at $x = 1$.

Solutions

For ii:

$$y = x\sqrt{5x^2 - 1}, \qquad x > \frac{1}{\sqrt{5}}.$$

For iii:

$$y = x\sqrt{6x^2 - 2x}, \qquad x > \frac{1}{3}.$$

Forming Differential Equations

Solution 102

In real-life models, a rate of change can depend on the current value. This extends the exponential growth studied in Year 1.

The population P of rabbits increases at a rate proportional to P. Form and solve a differential equation, with time t.

Form equation

$$\frac{dP}{dt} \propto P \quad \Rightarrow \quad \frac{dP}{dt} = kP.$$

Here $k > 0$ is the constant of proportionality.

Solve

For a positive population:

$$\frac{1}{P}\frac{dP}{dt} = k$$

$$\int \frac{1}{P}\,dP = \int k\,dt$$

$$\ln P = kt + C$$

$$P = e^{kt+C} = Ae^{kt}, \qquad A = e^C > 0.$$

Since $e^{kt} = (e^k)^t$, this can also be written $P = AB^t$, with $B = e^k$. An exponential model need not initially be expressed with base e.

Forming Differential Equations — Draining a Tank

Solution 103

A cylindrical tank has diameter 20 m. Water flows out at a rate proportional to the cube root of its volume V. Time t is in minutes and depth h is in metres.

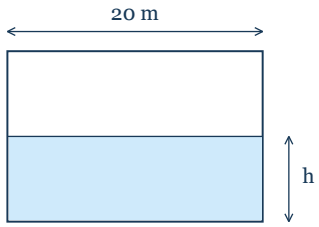
a. Show that $dh/dt = -k\sqrt[3]{h}$.

b. Find a general solution $h = (P - Qt)^{3/2}$.

Initially $h = 27$; after 10 minutes, $h = 8$.

c. Find P, Q.

d. Find the time when $h = 1$.



a

$V = \pi r^2 h = 100\pi h, \qquad \dfrac{dV}{dh} = 100\pi.$

$\dfrac{dV}{dt} = -c\sqrt[3]{V} = -c\sqrt[3]{100\pi h}.$

Outflow gives a negative rate. Using connected rates:

$\dfrac{dh}{dt} = \dfrac{dh}{dV}\dfrac{dV}{dt} = -\dfrac{c\sqrt[3]{100\pi}}{100\pi}\sqrt[3]{h} = -k\sqrt[3]{h}.$

Recall the chain rule from Chapter 9.

b

$h^{-1/3}\dfrac{dh}{dt} = -k \quad \Rightarrow \quad \int h^{-1/3}\,dh = \int -k\,dt.$

$\dfrac{3}{2}h^{2/3} = -kt + C \quad \Rightarrow \quad h^{2/3} = -\dfrac{2}{3}kt + \dfrac{2}{3}C.$

$h = \left(\dfrac{2}{3}C - \dfrac{2}{3}kt\right)^{3/2} = (P - Qt)^{3/2}.$

Use this until the tank becomes empty.

c

$27 = P^{3/2} \quad \Rightarrow \quad P = 9.$

$8 = (9 - 10Q)^{3/2} \quad \Rightarrow \quad 4 = 9 - 10Q.$

$Q = \dfrac{1}{2}.$

d

$1 = (9 - \tfrac{1}{2}t)^{3/2} \quad \Rightarrow \quad 1 = 9 - \tfrac{1}{2}t.$

$t = 16$ minutes.

Test Your Understanding — Inflow and Outflow

Solution 104

Edexcel C4 June 2005, Q8

Liquid enters an initially empty container at 20 cm^3/s. It leaves at a rate proportional to the volume V.

a. Explain why $dV/dt = 20 - kV$, with $k > 0$.

b. Solve to show $V = A + Be^{-kt}$, giving A, B in terms of k.

c. When $t = 5$, $dV/dt = 10$. Find V when $t = 10$.

a

The rate of increase is the rate in minus the rate out:

$\dfrac{dV}{dt} = 20 - kV.$

The outflow rate is kV; its contribution is negative.

b

$\displaystyle\int \frac{1}{20 - kV}\,dV = \int 1\,dt$

$-\dfrac{1}{k}\ln(20 - kV) = t + C.$

At $t = 0, V = 0$, $C = -\frac{1}{k}\ln 20$.

$t = \dfrac{1}{k}\ln\dfrac{20}{20 - kV}.$

$V = \dfrac{20}{k} - \dfrac{20}{k}e^{-kt}.$

$A = \dfrac{20}{k}, \qquad B = -\dfrac{20}{k}.$

c

$\dfrac{dV}{dt} = 20e^{-kt}.$

$10 = 20e^{-5k} \quad \Rightarrow \quad k = \dfrac{\ln 2}{5} \approx 0.139.$

$V(10) = \dfrac{100}{\ln 2}(1 - e^{-2\ln 2})$

$= \dfrac{100}{\ln 2}\left(1 - \dfrac{1}{4}\right) = \dfrac{75}{\ln 2} \approx 108 \text{ cm}^3.$

Exercise 11K — Pages 328–329 Solution 105

1. A rabbit population P satisfies $dP/dt = kP$, with $k > 0$ and time t in years. Initially there are 200 rabbits.

a. Find P in terms of k, t.

b. For $k = 3$, find when the population reaches 4000.

c. Explain a limitation of the model for large t.

1a

$$\int \frac{1}{P}\,dP = \int k\,dt \quad \Rightarrow \quad \ln P = kt + C.$$

$$t = 0, P = 200 \quad \Rightarrow \quad C = \ln 200.$$

$$\ln P - \ln 200 = kt \quad \Rightarrow \quad \ln \frac{P}{200} = kt.$$

$$\frac{P}{200} = e^{kt} \quad \Rightarrow \quad P = 200e^{kt}.$$

1b

$$4000 = 200e^{3t} \quad \Rightarrow \quad e^{3t} = 20.$$

$$t = \frac{\ln 20}{3} \approx 1.00 \text{ year}.$$

1c

Food and space are limited, so the population cannot continue to grow exponentially indefinitely.

Exercise 11K — Pages 328–329 Solution 106

2. The mass M of a plant's leaves varies with time according to $\dfrac{dM}{dt} = M - M^2$.

a. Given $M = 0.5$ at $t = 0$, find M in terms of t.

b. Find M when $t = \ln 2$.

c. Explain what happens to M as t increases.

Form equation

$$\int \frac{1}{M(1-M)}\,dM = \int 1\,dt.$$

$$\frac{1}{M(1-M)} = \frac{A}{M} + \frac{B}{1-M}.$$

$$1 = A(1-M) + BM.$$

$$M = 0: \ A = 1; \qquad M = 1: \ B = 1.$$

$$\int \left(\frac{1}{M} + \frac{1}{1-M}\right) dM = \int 1\,dt.$$

$$\ln|M| - \ln|1-M| = t + C.$$

$$\ln\left|\frac{M}{1-M}\right| = t + C.$$

$$\frac{M}{1-M} = Ke^t.$$

2a

$$\frac{0.5}{1-0.5} = Ke^0 \quad \Rightarrow \quad K = 1.$$

$$M = e^t(1-M) = e^t - e^t M.$$

$$M = \frac{e^t}{1+e^t}.$$

2b

$$t = \ln 2: \quad M = \frac{2}{1+2} = \frac{2}{3}.$$

2c

$$M = \frac{1}{e^{-t}+1}.$$

As t increases, $e^{-t} \to 0$, so $M \to 1$.

Exercise 11K — Pages 328–329 Solution 107

3. Ice on a pond has thickness x cm, increasing at a rate inversely proportional to x^2. Initially $x = 1$; after 20 days, $x = 2$.

a. Show that $x = \sqrt[3]{\frac{7}{20}t + 1}$.

b. Find the time for the thickness to increase from 2 cm to 3 cm.

3a

$$\frac{dx}{dt} \propto \frac{1}{x^2} \quad \Rightarrow \quad \frac{dx}{dt} = \frac{k}{x^2}.$$

$$\int x^2\,dx = \int k\,dt \quad \Rightarrow \quad \frac{x^3}{3} = kt + C.$$

$$t = 0, x = 1 \quad \Rightarrow \quad C = \frac{1}{3}.$$

$$x^3 = 3kt + 1.$$

$$t = 20, x = 2: \quad 8 = 60k + 1 \Rightarrow k = \frac{7}{60}.$$

$$x^3 = \frac{7}{20}t + 1 \quad \Rightarrow \quad x = \sqrt[3]{\frac{7}{20}t + 1}.$$

3b

When $x = 3$:

$$27 = \frac{7}{20}t + 1 \quad \Rightarrow \quad t = \frac{520}{7} \approx 74.3 \text{ days}.$$

The time from 2 cm to 3 cm is

$$\frac{520}{7} - 20 = \frac{380}{7} \approx 54.3 \text{ days}.$$

Exercise 11K — Pages 328–329 Solution 108

4. Tea at temperature $T\,^\circ$C cools in a room at 25 °C. Its rate of cooling is proportional to the difference between the tea and room temperatures.

a. Show that $dT/dt = -k(T - 25)$, explaining why $k > 0$.

Initially $T = 85$; 10 minutes later, $T = 55$.

b. Find the temperature after 15 minutes, to 1 decimal place.

4a

$$\frac{dT}{dt} \propto -(T - 25) \quad \Rightarrow \quad \frac{dT}{dt} = -k(T - 25).$$

The temperature difference is $T - 25 > 0$. The tea is cooling, so its derivative is negative. Thus k is positive; a negative k would make it warm.

4b

$$\int \frac{1}{T - 25}\,dT = \int -k\,dt$$

$$\ln|T - 25| = -kt + C.$$

$$t = 0, T = 85 \Rightarrow C = \ln 60.$$

$$\frac{T - 25}{60} = e^{-kt}.$$

$$t = 10, T = 55: \quad \frac{30}{60} = e^{-10k}.$$

$$\ln\frac{1}{2} = -10k \quad \Rightarrow \quad k = \frac{\ln 2}{10} \approx 0.0693.$$

At $t = 15$:

$$\frac{T - 25}{60} = e^{-15\ln 2/10}$$

$$T = 60e^{-15\ln 2/10} + 25 \approx 46.2\,^\circ\text{C}.$$

Exercise 11K — Pages 328–329 Solution 109

5. The surface area A mm^2 of an oil drop, at time t minutes, satisfies

$$\frac{dA}{dt} = \frac{A^{3/2}}{10t^2}.$$

At $t = 1$, $A = 1$.

a. Find A in terms of t.

b. Show that the area cannot exceed $400/361$ mm^2.

5a

$$\int A^{-3/2}\,dA = \int \frac{1}{10t^2}\,dt$$

$$-\frac{2}{\sqrt{A}} = -\frac{1}{10t} + C.$$

$$t = 1, A = 1: \quad -2 = -\frac{1}{10} + C \Rightarrow C = -\frac{19}{10}.$$

$$-\frac{2}{\sqrt{A}} = -\frac{1}{10t} - \frac{19}{10}.$$

$$\frac{2}{\sqrt{A}} = \frac{1}{10t} + \frac{19}{10} = \frac{1+19t}{10t}.$$

$$\sqrt{A} = \frac{20t}{1+19t}, \qquad A = \left(\frac{20t}{1+19t}\right)^2.$$

5b

For $t > 0$, $20t/(1+19t) < 20/19$.

As $t \to \infty$:

$$A \to \left(\frac{20}{19}\right)^2 = \frac{400}{361}$$

from below. This is the limiting upper bound, in mm^2.

Exercise 11K — Pages 328–329 Solution 110

6. A bath is modelled as a cuboid with base area 6000 cm^2. Water enters at 12000 cm^3/min and leaves through the plughole at $500h$ cm^3/min, where h is its depth in cm.

a. Show that $60\,dh/dt = 120 - 5h$, with time t in minutes.

Initially $h = 6$.

b. Find t when $h = 10$.

6a

$$V = 6000h \quad \Rightarrow \quad \frac{dV}{dh} = 6000.$$

$$\frac{dV}{dt} = 12000 - 500h.$$

The net rate is inflow minus outflow.

$$\frac{dh}{dt} = \frac{dh}{dV}\frac{dV}{dt} = \frac{1}{6000}(12000 - 500h)$$

$$= \frac{1}{60}(120 - 5h).$$

$$60\frac{dh}{dt} = 120 - 5h.$$

6b

$$\int \frac{60}{120 - 5h}\,dh = \int 1\,dt$$

$$-12\ln(120 - 5h) = t + C.$$

$$t = 0, h = 6: \quad C = -12\ln 90.$$

$$-12\ln(120 - 5h) = t - 12\ln 90.$$

When $h = 10$:

$$t = 12\ln 90 - 12\ln 70 = 12\ln\frac{9}{7} \text{ minutes.}$$

Exercise 11K — Pages 328–329

Solution 111

7a. Express $1/[P(10000-P)]$ in partial fractions.

A deer population P satisfies

$$\frac{dP}{dt} = \frac{1}{200}P(10000-P),$$

where t is in years. Initially $P = 2500$.

b. Solve, giving $P = a/(b+ce^{-50t})$.

c. Find the maximum population according to the model.

7a

$$\frac{1}{P(10000-P)} = \frac{A}{P} + \frac{B}{10000-P}.$$

$$1 = A(10000-P) + BP.$$

$$P=0 \Rightarrow A = \frac{1}{10000}; \quad P = 10000 \Rightarrow B = \frac{1}{10000}.$$

$$\frac{1}{P(10000-P)} = \frac{1}{10000}\left(\frac{1}{P} + \frac{1}{10000-P}\right).$$

7b

$$\frac{1}{10000}\int\left(\frac{1}{P} + \frac{1}{10000-P}\right)dP = \frac{t}{200} + C.$$

$$\frac{1}{10000}(\ln|P| - \ln|10000-P|) = \frac{t}{200} + C.$$

$$\ln P - \ln(10000-P) = 50t + D.$$

$$t=0, P=2500: \quad D = \ln\frac{2500}{7500} = \ln\frac{1}{3}.$$

$$\ln P - \ln(10000-P) - \ln\frac{1}{3} = 50t$$

$$\ln\frac{3P}{10000-P} = 50t \quad \Rightarrow \quad \frac{3P}{10000-P} = e^{50t}.$$

$$3P = (10000-P)e^{50t}$$

$$3Pe^{-50t} + P = 10000$$

$$P(3e^{-50t}+1) = 10000.$$

$$P = \frac{10000}{1+3e^{-50t}}.$$

$$a = 10000, \quad b = 1, \quad c = 3.$$

7c

Equilibria have $dP/dt = 0$, giving $P = 0$ or 10000. The population starting at 2500 approaches 10000 from below. This is its limiting maximum, not attained at finite time.

Exercise 11K — Pages 328–329

Solution 112

8. Liquid enters a container at 40 cm^3/s and leaks at $\frac{1}{4}V$ cm^3/s, where V cm^3 is its volume.

a. Show that $-4\,dV/dt = V - 160$.

Initially $V = 5000$.

b. Find $V = a + be^{-t/4}$, determining a, b.

c. Write down the limiting value of V as $t \to \infty$.

8a

$$\frac{dV}{dt} = 40 - \frac{1}{4}V \quad \Rightarrow \quad -4\frac{dV}{dt} = V - 160.$$

8b

$$\int\frac{-4}{V-160}\,dV = \int 1\,dt$$

$$-4\ln|V-160| = t + C.$$

$$t=0, V=5000: \quad C = -4\ln 4840.$$

$$4\ln 4840 - 4\ln|V-160| = t$$

$$\ln\left|\frac{4840}{V-160}\right| = \frac{t}{4}.$$

$$\frac{4840}{V-160} = e^{t/4} \quad \Rightarrow \quad \frac{V-160}{4840} = e^{-t/4}.$$

$$V = 160 + 4840e^{-t/4}.$$

$$a = 160, \qquad b = 4840.$$

8c

As $t \to \infty$, $e^{-t/4} \to 0$, so

$V \to 160$ cm^3.

Exercise 11K — Pages 328–329 Solution 113

9. In carbon dating, the amount R of radioactive carbon remaining in a fossil decreases at a rate proportional to R. Time t is in years.

a. Initially $R = R_0$. Show that $R = R_0 e^{-kt}$.

The half-life is 5730 years: after this time, half remains.

b. Find k exactly.

c. A fossil has 10% of its expected original carbon remaining. Find its age to the nearest year.

9a

$$\frac{dR}{dt} = -kR \quad \Rightarrow \quad \ln|R| = -kt + C.$$

$$t = 0, R = R_0: \quad C = \ln R_0.$$

$$\ln R - \ln R_0 = -kt \quad \Rightarrow \quad \frac{R}{R_0} = e^{-kt}.$$

$$R = R_0 e^{-kt}.$$

9b

$$t = 5730, R = \frac{R_0}{2}: \quad \frac{R_0}{2} = R_0 e^{-5730k}.$$

$$e^{5730k} = 2 \quad \Rightarrow \quad 5730k = \ln 2.$$

$$k = \frac{\ln 2}{5730}.$$

9c

$$\frac{R_0}{10} = R_0 e^{-kt} \quad \Rightarrow \quad e^{kt} = 10.$$

$$t = \frac{\ln 10}{k} = \frac{5730 \ln 10}{\ln 2} \approx 19035 \text{ years.}$$

Extension

Try STEP I 2011, Question 7(i) and (ii).

Summary of Functions Solution 114

Recall the method and integral. "Booklet" refers to the formula booklet used by the supplied course.

1. $\int \sin x \, dx$

Standard result.
$-\cos x + C$.
Booklet: No.

2. $\int \cos x \, dx$

Standard result.
$\sin x + C$.
Booklet: No.

3. $\int \tan x \, dx$

Use $\sin x / \cos x$, of the form $kf'(x)/f(x)$.
$\ln|\sec x| + C$.
Booklet: Yes.

4. $\int \sin^2 x \, dx$

Use $\cos 2x = 1 - 2\sin^2 x$:
$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$.
$\frac{1}{2}x - \frac{1}{4}\sin 2x + C$.
Booklet: No.

5. $\int \cos^2 x \, dx$

Use $\cos 2x = 2\cos^2 x - 1$:
$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$.
$\frac{1}{2}x + \frac{1}{4}\sin 2x + C$.
Booklet: No.

6. $\int \tan^2 x \, dx$

Use $1 + \tan^2 x = \sec^2 x$, so $\tan^2 x = \sec^2 x - 1$.
$\tan x - x + C$.
Booklet: No.

7. $\int \operatorname{cosec} x \, dx$

Substitute $u = \operatorname{cosec} x + \cot x$.
Derivation beyond this course's exam expectations.
$-\ln|\operatorname{cosec} x + \cot x| + C$.
Booklet: Yes.

8. $\int \sec x \, dx$

Substitute $u = \sec x + \tan x$.
Derivation beyond this course's exam expectations.
$\ln|\sec x + \tan x| + C$.
Booklet: Yes.

9. $\int \cot x \, dx$

Use $\cos x / \sin x$, of the form $f'(x)/f(x)$.
$\ln|\sin x| + C$.
Booklet: Yes.

Summary of Functions

Solution 115

Recall the method and integral. "Booklet" refers to the formula booklet used by the supplied course.

1. $\int \operatorname{cosec}^2 x \, dx$

   By observation.
   $-\cot x + C$.
   Booklet: No.

2. $\int \sec^2 x \, dx$

   By observation.
   $\tan x + C$.
   Booklet: Yes, but memorise.

3. $\int \cot^2 x \, dx$

   Use $1 + \cot^2 x = \operatorname{cosec}^2 x$.
   $-\cot x - x + C$.
   Booklet: No.

4. $\int \sin 2x \cos 2x \, dx$

   For sine and cosine with the same argument, use the double-angle identity:
   $\sin 2x \cos 2x = \frac{1}{2} \sin 4x$.
   $-\frac{1}{8} \cos 4x + C$.

5. $\int 1/x \, dx$

   Standard result.
   $\ln|x| + C$.
   Booklet: No.

6. $\int \ln x \, dx$

   Integration by parts: $u = \ln x$, $dv/dx = 1$.
   $x \ln x - x + C$.
   Booklet: No.

7. $\int \frac{x}{x+1} \, dx$

   Algebraic division:
   $x/(x+1) = 1 - 1/(x+1)$.
   $x - \ln|x+1| + C$.
   Booklet: No.

8. $\int \frac{1}{x(x+1)} \, dx$

   Partial fractions.
   $\ln|x| - \ln|x+1| + C$.
   Booklet: No.

Summary of Functions

Solution 116

1. $\int \frac{4x}{x^2+1} \, dx$

   Reverse chain rule: $kf'(x)/f(x)$.
   $2\ln|x^2+1| + C$.

2. $\int \frac{x}{(x^2+1)^2} \, dx$

   Consider $y = (x^2+1)^{-1}$ and differentiate, or substitute $u = x^2 + 1$.
   $-\frac{1}{2}(x^2+1)^{-1} + C$.

3. Find each integral:

   $$\int e^{2x+1} \, dx, \qquad \int \frac{1}{1-3x} \, dx.$$

   For a linear inner function, divide by the coefficient of x.
   $\frac{1}{2}e^{2x+1} + C$
   $-\frac{1}{3}\ln|1-3x| + C$.

4. $\int x\sqrt{2x+1} \, dx$

   Use integration by parts or a sensible substitution: $u = 2x+1$, or $u^2 = 2x+1$.
   $\frac{1}{15}(2x+1)^{3/2}(3x-1) + C$.

5. $\int \sin^5 x \cos x \, dx$

   Reverse chain rule.
   $\frac{1}{6}\sin^6 x + C$.